

A Review of Current Evidence for the Causal Impact of Attentional Bias on Fear and Anxiety

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Prominent cognitive theories postulate that an attentional bias toward threatening information contributes to the etiology, maintenance, or exacerbation of fear and anxiety. In this review, we investigate to what extent these causal claims are supported by sound empirical evidence. Although differences in attentional bias are associated with differences in fear and anxiety, this association does not emerge consistently. Moreover, there is only limited evidence that individual differences in attentional bias are related to individual differences in fear or anxiety. In line with a causal relation, some studies show that attentional bias precedes fear or anxiety in time. However, other studies show that fear and anxiety can precede the onset of attentional bias, suggesting circular or reciprocal causality. Importantly, a recent line of experimental research shows that changes in attentional bias can lead to changes in anxiety. Yet changes in fear and anxiety also lead to changes in attentional bias, which confirms that the relation between attentional bias and fear and anxiety is unlikely to be unidirectional. Finally, a similar causal relation between interpretation bias and anxiety has been documented. In sum, there is evidence in favor of causality, yet a strict unidirectional cause–effect model is unlikely to hold. The relation between attentional bias and fear and anxiety is best described as a bidirectional, maintaining, or mutually reinforcing relation.

Keywords: attentional bias, causality, fear, anxiety, phobia

Several epidemiological studies have shown that fear and anxiety disorders are highly prevalent in Western countries (e.g., Bijl, Ravelli, & Van Zessen, 1998; Kessler et al., 2005). According to

the World Health Organization (WHO), anxiety disorders are the most prevalent syndromes across different countries and cultures. Annual prevalence estimates vary between 2% and 5% in African and Asian countries, 5% and 12% in European and Latin American countries, and around 18% in the United States (WHO Mental Health Survey Consortium, 2004). Given the ubiquitous nature of fear and anxiety disorders, a thorough understanding of the causes of fear and anxiety and the identification of possible vulnerability factors for the development, maintenance, and relapse of anxiety disorders are imperative.

According to information-processing theories, fear and anxiety may be caused by different cognitive processes, such as interpretation, memory, and attention (e.g., Beck & Clark, 1997; Eysenck, 1992, 1997; Mogg & Bradley, 1998; Williams, Watts, MacLeod, & Mathews, 1988, 1997). Anxious individuals are hypothesized to display biases in these cognitive processes. Compared to nonanxious individuals, anxious individuals are assumed to be more likely to interpret a neutral or ambiguous stimulus as being threatening (interpretation bias), to more easily recall threatening events from memory (memory bias), and to show a preference to attend to threatening stimuli over nonthreatening stimuli in their environment (attentional bias).

Several theoretical positions have been proposed about the nature of the relation between attentional bias and fear and anxiety. Some authors have made strong causal claims, on both theoretical and more empirical grounds. For instance, Beck and Clark (1997,

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p. 49) stated that “A core tenant (*sic*) of these [cognitive] theories is that the type of emotional information and the manner in which it is processed are crucial factors in the etiology, maintenance, and treatment of anxiety disorders.” Mathews and MacLeod (2002, p. 333) argued that “it is the type of processing style, or bias, that is elicited by events which causes vulnerability to anxiety, instead of biased processing being only an incidental by-product of emotional variations.” In their model of social anxiety, Rapee and Heimberg (1997) stated that “distortions and biases in the processing of social/evaluative information lead to heightened anxiety in social situations and, in turn, help to maintain social phobia” (p. 741). Such causal views are in contrast with the idea that attentional bias is a mere consequence or an epiphenomenon of fear and anxiety (Beck, Emery, & Greenberg, 1985).

In line with growing scientific interest in cognitive biases in fear, anxiety, and anxiety disorders, several literature reviews on this topic have been written (see, e.g., Buckley, Blanchard, & Neill, 2000; Cisler, Bacon, & Williams, 2009; Cisler & Koster, 2010; Heinrichs & Hofmann, 2001; Mathews & MacLeod, 2005; Mobini & Grant, 2007; Puliafico & Kendall, 2006; Yiend, 2010). As a general rule, the authors of these reviews conclude that there is ample evidence for the existence of cognitive biases in fear and anxiety. However, the thorny question of causality has thus far largely been neglected. This question is nonetheless crucial, not only from a scientific perspective but also from a clinical perspective. If cognitive biases are causally involved in the development or maintenance of fear and anxiety, then therapeutic interventions should aim at reducing these cognitive biases to prevent or reduce the individual’s level of fear or anxiety.

Our present review is unique in that we organize the broad literature on attentional bias to provide a comprehensive overview of the evidence concerning the causal nature of the relation between attentional bias and fear and anxiety. To achieve this aim, we adopt a broad perspective on the concepts of “fear,” “anxiety,” and “causality.” With regard to the concepts of fear and anxiety, we base our description of specific findings on the terminology of Rachman (1998). We consider fear a specific and immediate emotional reaction to a specific and well-defined class of stimuli or situations, such as spiders or needles. Anxiety is the more general, enduring, and vague feeling of unease and stress, characterizing high-trait-anxious individuals. We use the words *high anxious* or *high fearful* to describe individuals with elevated scores on anxiety or fear measures, without these individuals being diagnosed as clinically anxious. The notion of emotional distress is closely linked to anxiety, and we use it to refer to a compound of feelings of anxiety, stress, and negative mood. Stress reactivity or stress responsiveness typically denotes the feelings of anxiety or fear in response to an external stressor or stressful event (e.g., a public speech task). Finally, we use the terms *anxiety disorder* or *clinical anxiety* to refer to fear and anxiety of clinical severity, as described in the *Diagnostic and Statistical Manual of Mental Disorders* (4th ed., text rev.; DSM-IV-TR; American Psychiatric Association, 2000).

In a similar vein, we adopt a broad view on causality. We do not restrict causality to a strict unidirectional cause–effect model in which attentional bias is involved in the etiology of fear and anxiety disorders. Rather, we also take into consideration the possibility that attentional bias is a vulnerability factor for anxiety disorders. A vulnerability factor may increase the

chance of a certain outcome, without necessarily determining the outcome in isolation (Kraemer et al., 1997; Kraemer, Stice, Kazdin, Offord, & Kupfer, 2001). Attentional bias toward threat may increase the chance of developing fear or anxiety disorders, without being a necessary or sufficient cause of fear or anxiety. Finally, we also consider the possibility that attentional bias toward threat is a maintaining or exacerbating factor for fear or anxiety.

Our view on causality can be clarified if one thinks of a man falling through the air (note that we consider only the actual falling, not the possibly fatal consequence of the falling). From a strict unidirectional cause–effect view on causality, being pushed out of a plane can be seen as a cause of the man being in the air; being pushed out of the plane is the main factor involved in the etiology of being in the air. However, the man can only be pushed out of the plane if he got onto the plane. Hence, being on a plane constitutes a vulnerability factor for falling through the air. Such vulnerability factors are often neither necessary nor sufficient for an event to happen. The man could also be pushed off a high building or out of a hot air balloon, and not everyone who enters a plane will be pushed out. However, vulnerability factors increase the chance of an event to happen. It is clear that the man cannot fall through the air if he stays on the ground. Next, consider the case in which the man wears a parachute and drifts into a thermal column. The parachute and the hot air will prolong the time that the man is in the air. Hence, wearing a parachute and drifting into thermals are maintaining or exacerbating factors of being in the air, without being causal factors for being in the air initially.

Our review also has boundaries. First, we focus on the role of attentional bias in fear and anxiety, and we do not elaborate on other cognitive biases (but see the Analogy section for a concise discussion of this topic). Second, although it is possible that some findings are different in specific anxiety problems of specific severity, we do not explicitly go into detail on these matters. This is mainly because many cognitive theories of fear and anxiety (see below) assume that the same processes underlie different anxiety problems of differing intensity. Third, we do not review every piece of evidence that might be relevant for the topic of our article. Over the past decades, a wealth of studies on attentional bias within the context of fear and anxiety disorders has been published. Our aim is to give the reader a comprehensive overview of current evidence concerning the causal nature of the relation between attentional bias and fear and anxiety, and to highlight the most important findings, possible limitations, boundary conditions, and future perspectives.

Our review consists of five parts. First, we discuss the most prominent models and theories that postulate a causal relation between attentional bias and fear or anxiety. Second, we briefly explain the most common measures used in attentional bias research. Third, we introduce Hill’s (1965) criteria of causation. Fourth, we use these criteria to structure the available knowledge about the relation between attentional bias and fear and anxiety. Fifth, we conclude that there is indeed evidence in favor of causality, and we consider perspectives and challenges for future research.

Cognitive Theories of Fear and Anxiety

The Schema-Based Theory of Beck and Clark (1997)

Beck and Clark (1997) presented an influential schema-based model of information processing, according to which biases in three stages of information processing lie at the core of anxiety. In the first stage, stimuli are automatically processed, and their valence and personal relevance is assessed. In the second phase, a primal threat mode becomes activated if a stimulus has been evaluated as threatening in the first stage. This primal threat mode dominates information processing, resulting in several cognitive biases such as attending selectively to threat cues. The output of this second phase is a complex pattern of behavioral, cognitive, physiological, and affective responses, which Beck and Clark refer to as a state of "anxiety." Finally, if in a third stage the primal threat mode is hyperactive, it will block the activation of more constructive reappraisal processes and anxiety will further increase. According to Beck and Clark, effective treatment of anxiety should aim at both weakening of the automatic activation of the primal threat mode (i.e., the origin of anxiety) and strengthening more constructive reappraisal (i.e., decreasing perceived threat and increasing perceived coping potential).

Beck and Clark (1997) are rather vague about the causal nature of the relation between attentional bias and anxiety (see Table 1). First, their model is explicitly based on the model of Beck et al. (1985), and they refer to Beck et al. when making the relatively strong causal claim on page 49 that the type of information processing is involved in the etiology or maintenance of anxiety disorders. However, Beck et al. argued *against* the notion that attentional or other cognitive biases are the main cause of psychopathology, a notion that they found "just as illogical as an assertion that hallucinations cause schizophrenia" (p. 85). Second, they reason that activation of the primal threat mode leads to cognitive biases (p. 53). On page 57, they argue that the result of the activation of the primal threat mode is an "automatic behavioral/physiological/affective/cognitive goal-directed response pattern" that is called "anxiety." From this phrasing, it is unclear whether (a) cognitive biases and anxiety are independent effects of the activation of the primal threat mode, (b) cognitive biases are part of the automatic cognitive response pattern called anxiety, or (c)

cognitive biases result from the primal threat mode and *in turn* lead to anxiety. Of these three possible interpretations, only the last one contains a notion of causality.

The Information-Processing Model of Williams et al. (1988, 1997)

According to Williams et al. (1988, 1997), a preattentive affective decision mechanism assesses the threat value of all incoming stimuli. High levels of state anxiety increase the perceived threat level of stimuli as assessed by the affective decision mechanism. Depending on the level of threat, the output of the affective decision mechanism is forwarded to a cognitive resource allocation mechanism. Trait anxiety reflects a proneness to direct cognitive resources either toward or away from threatening stimuli. High-trait-anxious people will direct their attention toward threatening stimuli, whereas low-trait-anxious people will direct processing resources away from threat.

Importantly, Williams et al. (1988, 1997) proposed that high-trait-anxious individuals will allocate even more cognitive resources toward threatening stimuli when they are in a stressful situation (i.e., when state anxiety is high). Such a tendency to allocate more cognitive resources to threat is argued to increase "the likelihood that such [high-trait-anxious] people will exhibit greater psychopathology in response to smaller degrees of affective disturbance" (Williams et al., 1997, p. 309). As such, Williams et al. regard attentional bias as a *vulnerability factor* for the development of anxiety disorders within a diathesis-stress framework (see Table 1). From this perspective, clinical anxiety is determined by the interaction between individual vulnerabilities and the environment. If the vulnerable individual is confronted with high levels of stress or state anxiety, he or she will be more likely to become clinically anxious. The tendency to allocate cognitive resources to threat is also argued to prolong episodes of emotional disturbance and delay recovery from emotional disturbance. Hence, Williams et al. (1997) suggested that attentional bias maintains elevated anxiety levels when they state that "the effect of such tendencies [i.e., attentional bias] would be to prolong any episode of emotional disturbance" (p. 309).

Table 1
Summary of Causality Assumptions of Different Theories

Theory	Attentional bias causes fear and anxiety in a unidirectional cause-effect model	Attentional bias is a cognitive vulnerability factor for the development of fear and anxiety	Attentional bias is a maintaining or exacerbating factor for fear and anxiety
The cognitive view of Eysenck (1992, 1997)	No	Yes	Yes
The cognitive-motivational analysis of Mogg and Bradley (1998)	No	No	Yes (vigilance followed by subsequent avoidance)
The information-processing model of Williams et al. (1988, 1997)	No	Yes	Yes
The integrative model of Bar-Haim et al. (2007)	Yes	Yes	Not specified
The schema-based theory of Beck and Clark (1997)	Yes (if attentional bias is considered a link in the chain leading to fear and anxiety)	No	No

Eysenck's (1992, 1997) Cognitive View on Anxiety

The basic tenet of Eysenck's (1992) theorizing is that cognitive vulnerability factors play a central role in the etiology of anxiety disorders (see Table 1). In his theory, the most important cognitive characteristic of anxiety is hypervigilance. The construct hypervigilance is similar to attentional bias as we defined it, but in Eysenck's theorizing it has a broader meaning. It encompasses not only a preference to attend to threatening stimuli over nonthreatening stimuli (attentional bias) but also increased general distractibility, a high rate of environmental scanning, a broad attentional window prior to the detection of threat, and a narrow or focused attentional window after threat has been detected. Eysenck conceptualizes hypervigilance as a latent vulnerability factor for the development of anxiety disorders. Hence, anxiety disorders will develop if the cognitively vulnerable individual is exposed to stressful life events (i.e., a diathesis-stress model). Because hypervigilant individuals have a higher likelihood of detecting threatening stimuli in their environment, these individuals will perceive the environment as more threatening, which in turn increases the chance that they will develop a clinical anxiety disorder.

Eysenck (1997) further elaborated these ideas. In line with his initial theory, attentional bias is argued to be a cognitive vulnerability factor for the development of anxiety disorders within a diathesis-stress model. However, Eysenck subsequently argued that "clinical anxiety develops when there is a positive feedback loop between these biases and state anxiety, in which high levels of anxiety exaggerate the biases, and the exaggerated biases increase state anxiety" (p. 119). This formulation suggests that attentional bias is regarded as a maintaining or exacerbating factor for anxiety disorders, with higher levels of anxiety leading to more attentional bias and more attentional bias leading to higher levels of anxiety.

The Cognitive-Motivational Analysis of Anxiety of Mogg and Bradley (1998)

Like the model of Williams et al. (1997), the cognitive-motivational view identifies two functional systems that are implicated in attentional bias: a valence evaluation system and a goal engagement system (Mogg & Bradley, 1998). The threat value of a stimulus is assessed by the valence evaluation system. The outcome of the valence evaluation system is passed on to the goal engagement system. If a stimulus is evaluated as threatening, the goal engagement system will interrupt the pursuit of current goals and prioritize the processing of the threatening stimulus. Hence, the allocation of attention is determined by the outcome of the valence evaluation system. The default reactivity to threat in the valence evaluation system is determined by the level of trait anxiety of the subject. If trait anxiety is high, the valence evaluation system is more likely to tag a stimulus as threatening, even when the objective threat value of the stimulus is mild. Hence, whereas all individuals attend to highly threatening stimuli, only high-trait-anxious individuals attend to mild threat.

According to the cognitive-motivational view, an attentional bias toward mildly threatening stimuli is a sign of anxiety vulnerability without necessarily playing a crucial role in the etiology of anxiety. The model postulates that hypersensitivity of the valence evaluation system is the primary causal factor for anxiety vulner-

ability and for increased attentional bias to mild threat. However, an attentional bias to mild threat is regarded as a potential maintaining factor of clinical anxiety (see Table 1). If initial orienting to threat is followed by avoidance, habituation will be hampered and anxiety is maintained. Furthermore, for high-trait-anxious individuals, even mildly threatening stimuli will interrupt the pursuit of ongoing goals and enter the focus of attention. Consequently, these individuals will be especially likely to perceive their environment as threatening, which may in turn increase state anxiety.

The Integrative Model of Bar-Haim, Lamy, Pergamin, Bakermans-Kranenburg, and van IJzendoorn (2007)

Bar-Haim et al. (2007) proposed an integrative model that combines several aspects of the models of Williams et al. (1988, 1997) and Mogg and Bradley (1998). The model comprises four systems. The threat value of incoming stimuli is assessed in the preattentive threat evaluation system. If threat is high, a resource allocation system is activated, which in turn results in physiological alertness, the interruption of ongoing activity, and attentional orienting to the location of the stimulus. The outcome of the resource allocation system serves as input for a guided threat evaluation system, in which the current stimulus is compared with memory and prior learning experiences, and in which the context and possible coping mechanisms are taken into consideration. The outcome of this guided threat evaluation system is a conscious evaluation of the threat value of the stimulus. If the consciously evaluated threat level is high, a goal engagement system will interrupt the pursuit of current goals and the primary goal of the individual will be to alleviate anxiety.

According to the model, attentional bias results from the operation of the resource allocation system. More specifically, a hypersensitive resource allocation system will result in the allocation of attentional resources even to stimuli that are evaluated as only mildly threatening by the preattentive threat evaluation system. This biased resource allocation system is argued to be a cause of anxiety problems (see Table 1), as Bar-Haim et al. (2007) state that "high-trait anxiety or different anxiety disorders may stem from . . . a bias in the RAS [resource allocation system], that is, a tendency to allocate resources even to stimuli evaluated as only mildly threatening" (p. 17). This phrasing implies that attentional bias may be involved in the *etiology* of anxiety disorders (a strict cause-effect relation), or can at least be considered a vulnerability factor for the development of clinical anxiety.

Causality in Cognitive Models of Fear and Anxiety: Conclusions

Although all five theories discussed above consider attentional bias to threatening stimuli a key characteristic of fear and anxiety, the models differ with regard to the causal nature of the relation between attentional bias and fear or anxiety. Table 1 illustrates that the extent to which different cognitive theories of fear and anxiety postulate a causal relation between biased attention and clinical anxiety depends largely on how one defines causality. As we have argued before, causality not only refers to strict linear cause-effect relations (etiology), but also is involved in the impact of vulnerability factors, maintaining factors, and exacerbating factors. Only

the model of Bar-Haim et al. (2007) adopts a strict linear interpretation (A leads to B) of the relation between attentional bias and fear or anxiety. According to our broad view on causality, however, all five theories assume some causal influence of attentional bias on clinical anxiety. To assess as well the evidence for a bidirectional, mutually reinforcing, or maintaining relation, we consider not only evidence in favor of the idea that attentional bias causes fear and anxiety, but also evidence in favor of the idea that fear and anxiety causally influence attentional bias. In the main part of this review, we examine to what extent these hypothetical causal relations are supported by empirical research. However, before doing so, we give a concise overview of the measures that have been used in research on attentional bias.

Common Measures of Attentional Bias

A wide variety of different tasks have been developed to study biased attention.¹ An extensive discussion of all possible paradigms is not within the scope of this article (for a more elaborate review of measures of attention, see Weierich, Treat, & Hollingworth, 2008). We provide only a brief description of the tasks that have most often been used to assess attentional bias, with the aim to facilitate the literature review that we present in the remainder of our review (see Table 2). Some of these tasks have been used for other purposes as well, but our description focuses on the assessment of biased attention for threat.

The Emotional Stroop Task

In the emotional Stroop task, words with varying threat value are presented in different colors (for a review, see Williams, Mathews, & MacLeod, 1996). Participants are required to respond as fast as possible to the color of the word, while ignoring the meaning of the word. Biased attention for threat is inferred when color naming is slower or less accurate for threatening words than for nonthreatening words. The common interpretation of the emotional Stroop effect is that the threatening meaning of the word draws attention, leaving less attentional resources for the naming of the color.

However, this interpretation has been disputed. For instance, de Ruiter and Brosschot (1994) argued that the emotional Stroop effect could also reflect cognitive avoidance rather than attentional bias. Furthermore, MacLeod, Mathews, and Tata (1986) suggested that the processing of negative words increases state anxiety and hence slows down reaction times (see also Algorn, Chajut, & Lev, 2004; De Houwer, 2003; Larsen, Mercer, & Balota, 2006). This issue aside, the psychometric properties of the emotional Stroop task are problematic. Several authors have reported low test–retest reliability for the emotional interference effect (i.e., the difference between color-naming latencies of neutral and emotional words) in the emotional Stroop task (e.g., Eide, Kemp, Silberstein, Nathan, & Stough, 2002; Kindt, Bierman, & Brosschot, 1996; Strauss, Allen, Jorgensen, & Cramer, 2005; but see Dresler, Mériaux, Heekeren, & van der Meer, 2009).

The Dot Probe Task

In the dot probe task (MacLeod et al., 1986), two cue stimuli appear simultaneously at different spatial locations on a computer

screen. One of these cues is threatening, whereas the other cue is not. After a short interval, the cues disappear and a target stimulus replaces one of the two cues. Participants are required to respond as fast and as accurately as possible to the location or the identity of the target. An attentional bias toward threatening stimuli is inferred from faster reaction times on trials where the target appears at the same spatial location as the threatening cue (congruent trials) than on trials where the target appears at the location of the nonthreatening cue (incongruent trials). Hence, attentional bias in the dot probe task could arise from fast responding on congruent trials (attentional engagement to threat), by slow responding on incongruent trials (slow attentional disengagement away from threat), or a combination of both (e.g., Koster, Crombez, Verschuere, & De Houwer, 2004).

The reliability of the dot probe task was systematically assessed by Schmukle (2005; see also Cooper et al., 2011; Van Bockstaele et al., 2011a). He found that neither a pictorial nor a verbal version of the task had good split-half reliability or good test–retest reliability. In contrast, Bar-Haim et al. (2010) found a notably high split-half reliability in the dot probe task.

The Emotional Spatial Cueing Paradigm

In the emotional adaptation of the spatial cueing task (Fox, Russo, Bowles, & Dutton, 2001), a single cue stimulus, either threatening or nonthreatening, is presented in one of two possible locations. After a brief interval, the cue disappears and a target appears either in the previously cued location (valid trials) or in the opposite location (invalid trials). Participants are required to respond as fast and as accurately as possible to the identity or the location of the target. A cue validity index is computed by subtracting reaction times on valid trials from reaction times on invalid trials. Attentional bias toward threat is inferred from greater cue validity indices on trials with a threatening cue compared to trials with a nonthreatening cue. As in the dot probe task, attentional bias can be caused by faster responding on valid trials, by slower responding on invalid trials, or by a combination of both factors. To our knowledge, the reliability of the exogenous cueing task has not been assessed.

The Visual Search Task

In a visual search task (e.g., Öhman, Flykt, & Esteves, 2001), participants are required to find and respond to a target stimulus that is embedded within an array of distracting stimuli. In some versions of the task, the threat value of the target is manipulated, whereas the distractors are neutral. In such cases, attentional bias is inferred from faster reaction times on trials with threatening targets compared to trials with neutral targets. In other versions, the threat value of the distractors varies over trials, whereas the target is always neutral. Attentional bias is then derived from slower reaction times on trials with threatening distractors com-

¹ In this review, we limit ourselves to the most often used behavioral measures of attention. Although eye movements have also been used as an indicator of attentional deployment (e.g., Hermans, Vansteenwegen, & Eelen, 1999), we believe that there is little agreement or consistency as to how attentional bias is defined in such studies (e.g., total dwell time on threat, percentage of first fixations on threat, speed of first fixation on threat, or duration of threat fixations).

Table 2
Summary of Attentional Bias Measures

Measure	Task	Index of attentional bias	Reliability	Convergent validity
Emotional Stroop	Name color of threatening and neutral words	Slower color naming (more interference) for threatening words than for nonthreatening words	Low	Low (with dot probe)
Dot probe	Two cues (threatening and neutral), respond to target appearing on threatening (congruent) or neutral (incongruent) location	Faster reaction times on congruent than on incongruent trials	Low	Low (with Stroop)
Exogenous cueing	Single cue (threatening or neutral), respond to target appearing on cued (valid) or noncued (invalid) location	Difference in reaction times between valid and invalid trials is larger for trials with threatening cues than for trials with neutral cues	Unknown	Unknown
Visual search	Find threatening target in array of neutral distractors or find neutral target in array of threatening distractors	Faster responses on trials with threatening target; slower responses on trials with threatening distractors	Unknown	Unknown
Attentional blink	Identify two targets (T1 and T2, one of which is threatening) in a stream of distractors	Deterioration of T2 identification due to short lag between T1 and T2 is smaller when T1 is neutral and T2 is threatening, but larger when T1 is threatening and T2 is neutral, compared to neutral T1 and T2	Low	Unknown

Note. Our assessment of reliability was based on correlations between two test moments (test–retest reliability), correlations between two subsets of data (split-half reliability), or a combination of both. Our assessment of convergent validity was based on the correlations between indices of attentional bias as measured with different tasks. In line with Cohen (1992), we considered (the absolute value of) correlations that were in the range of .10, .30, and .50 as small, medium, and large, respectively. Hence, our assessment of the reliability and validity of the attentional bias measures as “low” was based on the fact that the large majority of the relevant correlations in the literature were in the range of .20 or lower. T1 = first target; T2 = second target.

pared to trials with neutral distractors. To our knowledge, the reliability of the visual search task has not been assessed.

The Attentional Blink Task

In the attentional blink task, participants are shown a rapid serial visual presentation stream of distractors (Raymond, Shapiro, & Arnell, 1992). Two target stimuli (T1 and T2) are embedded within this stream of stimuli. At the end of the stream, participants are required to report these target stimuli. Typically, performance for T1 is good. However, performance for T2 depends on the temporal lag between T1 and T2. When this lag is short (i.e., between 200 and 400 ms), T2 performance is hampered relative to when the lag is long (i.e., >400 ms). The common interpretation for this finding is that the identification of T1 consumes limited attentional resources, leaving insufficient resources for the identification of T2 at short lags.

In most emotional versions of this task, the threat level of one of the two target stimuli is manipulated (see Yiend, 2010). One method is to manipulate the threat value of T2, while keeping T1 neutral. The rationale is that arousing/threatening T2 stimuli are processed more efficiently and thereby show a diminished attentional blink (e.g., Anderson, 2005; Keil & Ihssen, 2004). In this case, attentional bias toward threatening T2 stimuli is thus inferred from a smaller attentional blink effect with threatening T2 stimuli relative to neutral T2 stimuli. A second method is to manipulate the arousal/threat value of T1, while keeping T2 neutral. Participants are assumed to have problems disengaging their attention away from threatening T1 stimuli. Therefore, the attentional blink should increase more after the presentation of an arousing/threatening T1 stimulus than after a neutral T1 stimulus (e.g., Ihssen &

Keil, 2009; Mathewson, Arnell, & Mansfield, 2008). In such designs, an attentional bias toward threatening stimuli is inferred from a larger attentional blink effect on trials with threatening T1 stimuli relative to trials with neutral T1 stimuli. In a third version of this task, only one target is presented, but it is preceded by either a neutral or an arousing/threatening distractor. Results show that arousing distractors interfere with target processing when the lag between the crucial distractor and the target is short, but not when it is long. This effect is called emotion-induced blindness or attentional rubbernecking (Arnell, Killman, & Fijavz, 2007; Most, Chun, Widders, & Zald, 2005).

To our knowledge, the reliability of the attentional blink task to capture attentional bias to threat has not been assessed. However, Tibboel, De Houwer, and Field (2010) found that the split-half reliability of an alcohol-related attentional blink was very low.

Measures of Attentional Bias: Discussion

Although all paradigms discussed above are frequently used to measure attentional bias in fear and anxiety, two issues need to be taken into consideration. First, the paradigms differ with regard to how attention is measured (see Table 2). The emotional Stroop task and the attentional blink task measure attention to the semantic content of stimuli. In these tasks, attentional bias is inferred from differences in reaction times or accuracy between two semantically different classes of stimuli. The attentional blink task further has a temporal component, as different stimuli are presented in a rapid stream. In the dot probe task, the exogenous cueing task, and the visual search task, there is also a spatial component. In these paradigms, attentional bias is inferred from a difference in reaction time toward semantically different classes of

stimuli that can appear in different spatial locations. Several authors have tried to differentiate further between different components of attention (engagement, disengagement, and shifting; see Posner & Petersen, 1990) in the dot probe task (Koster et al., 2004; Koster, Crombez, Verschuere, & De Houwer, 2006; Salemink, van den Hout, & Kindt, 2007a), the emotional exogenous cueing task (Fox et al., 2001; see also Fox, Russo, & Dutton, 2002; Van Damme, Crombez, Hermans, Koster, & Eccleston, 2006), and the visual search task (e.g., Derakshan & Koster, 2010; Matsumoto, 2010; Waters, Lipp, & Spence, 2008). However, the calculation or operationalization of these different components of attention has been disputed (Mogg, Holmes, Garner, & Bradley, 2008; but see Cisler & Olatunji, 2010). Although the measurement of the separate components has been challenged and current research is inconclusive, there is general agreement that the dot probe task, the exogenous cueing task, and the visual search task are useful measures of attentional bias as a single entity that incorporates these components. Because the focus of our review is on attentional bias as a single entity, we do not elaborate on the different components of attention.

Second, although all measures that we discussed above are frequently used in research on attentional bias in fear and anxiety, many of these paradigms have limitations. In our opinion, the most important limitation concerns the psychometric properties of the paradigms. As we argued earlier, the reliability of these tasks is either poor or unknown. Another notable psychometric weakness of the attentional bias measures is their low degree of convergent validity. Although they are all assumed to measure the same theoretical construct, different measures of attentional bias often do not correlate with one another. For instance, Mogg et al. (2000) found near zero correlations between attentional bias toward threat in a dot probe task and emotional Stroop interference scores (for highly similar outcome, see Dalgleish et al., 2003). Salemink et al. (2007a) found a notably small correlation of .10 between attentional bias scores as measured in two versions of the dot probe task (target identification vs. target localization) within a single sample. Although discouraging, such low levels of convergence are unsurprising given the low reliability of the measures. It is clear that these poor psychometric properties are undesirable and that they complicate the interpretation of individual findings and the larger picture of the entire field. A major scientific effort is needed to meticulously assess and improve the psychometric properties of measures of attentional bias, or even to construct entirely new paradigms. Without such effort, progress in the field of attentional bias in fear and anxiety is bound to be equivocal.

Hill's (1965) Criteria of Causation

Several authors have formulated sets of criteria for inferring a causal relation between two variables (e.g., Lilienfeld, 1959; MacMahon & Pugh, 1970; Susser, 1973). Probably the most influential and most extensive set of criteria has been formulated by Sir Austin Bradford Hill (1965) for epidemiological research in medical settings and public health. He proposed that the likelihood of a causal relation between two variables can be evaluated on the basis of nine criteria. These nine criteria are (a) the strength of the association, (b) consistency, (c) specificity, (d) a dose-response curve, (e) plausibility, (f) coherence, (g) temporality, (h) experimental evidence, and (i) analogy. In our review of the literature,

we used several of these criteria as a heuristic tool for selecting and organizing relevant studies. The current section therefore contains a brief description of these criteria and their relation to the issue of causality.

With regard to the *strength of the relation*, it is clear that for A to cause B, at the minimum, there must be a relation between A and B. In other words, the probability that B is present should depend on the presence of A. Although Hill (1965) argued that stronger effects are more likely to be causal, he also admitted that relatively weak effects should not be dismissed as being noncausal. Furthermore, the presence alone of a relation is not sufficient to conclude that A causes B. It is possible that B causes A, or even that a third factor C causes both A and B. *Consistency* refers to the replicability and generalizability of findings. For instance, one can examine whether an association has been observed in different laboratories and countries, using different methodologies. *Specificity* refers to the presence of a one-to-one relation between A and B. In other words, A only leads to B, and B is only caused by A. The criterion of *dose-response curve* requires a continuous and monotonic relation between A and B. The higher someone's score for A, the higher this person should score for B. In other words, the individual level in A should be related to the individual level in B. Dose-response curves are expressed in terms of correlations between continuous variables. It should be clear that correlations do not imply causation. However, many causal relations give rise to significant correlations. With *plausibility*, Hill indicated that the mechanism between cause and effect must make sense in the light of current theories and results. Of course, what can be considered plausible depends largely on the scientific knowledge of the day. A sixth criterion is *coherence*, which implies that the suggested causal relation between A and B cannot be in conflict with other existing knowledge about A, B, or the relation between A and B.

The criteria representing the most convincing evidence for causality are temporality and experimental evidence. With *temporality*, Hill (1965) indicated that for A to cause B, A must precede B in time. Typically, prospective studies can provide evidence for temporality. *Experimental studies* are crucial for the demonstration of causality. If one can show that experimentally induced changes in A lead to changes in B, this constitutes strong evidence for causality. However, even experimental evidence should be interpreted with caution. It is still possible that manipulations of A lead to changes in an unidentified process C, which in turn lead to changes in B. Hence, if the relation between A and B is indirectly causal (i.e., $A \rightarrow C \rightarrow B$), it might be more interesting to identify and experimentally manipulate C. For example, imagine a country where smoking is legal in public places and no one has ever investigated the effect of smoking on public health. One could formulate the hypothesis that spending time in bars causes lung cancer. An experimental study in which the incidence of lung cancer in participants who are randomly restrained from visiting bars is compared with the incidence of lung cancer in participants who still visit bars may easily show higher prevalence rates of lung cancer in the latter group of participants. Hence, as long as smoking has not been identified as the real cause of lung cancer, researchers may believe that spending time in bars is the cause of lung cancer. This example clearly illustrates that findings obtained through experimental manipulation can be misinterpreted and that the outcome of every experiment may be explained in several ways. Another limitation of experimental studies involving hu-

mans is that results might be due to demand effects, placebo effects, or Hawthorne effects (i.e., participants behave differently just because they know that they are being studied; Adair, 1984). These confounding factors must be carefully addressed within experimental designs, especially by including proper control conditions and randomizing group assignment. The last criterion concerns *analogies*. If there are examples in scientific research describing a causal relationship between the constructs A' and B', then a similar causal relation between A and B is also more plausible.

Over the past 45 years, Hill's (1965) criteria have been widely cited and have been used to assess causality in a variety of domains, including medical science (e.g., Bosch, Lorincz, Muñoz, Meijer, & Shah, 2002; Perrio, Voss, & Shakir, 2007), neuroscience (e.g., van Reekum, Streiner, & Conn, 2001), psychiatry (e.g., Thornicroft, 1990) and clinical psychology (e.g., Birmingham, Touyz, & Harbottle, 2009). However, we have omitted two of Hill's original nine criteria, because—for different reasons—these two criteria are less suitable in our present review.

First, as Hill (1965) noted himself, many effects may have multiple causes, so although specificity is an indication of causality, the absence of specificity is by no means an argument against causation. Most psychologists nowadays would agree that the causal pathways of fear and anxiety disorders probably consist of a complex interplay between genes and environment (e.g., Craske et al., 2009; Eysenck, 1997; Merckelbach, de Jong, Muris, & van den Hout, 1996; Williams et al., 1997), rendering the specificity criterion less relevant.

Second, we refrain from discussing the coherence criterion in a separate section. This is not because we feel that coherence is a less valuable criterion. In critically reviewing the evidence for any scientific hypothesis or theory, it is imperative to take into account evidence that is compatible with the hypothesis as well as evidence that is incompatible with the hypothesis. However, rather than review incompatible evidence in a separate section, we have integrated this evidence in our discussion of the remaining seven criteria.

Finally, it is important to note that Hill (1965) never meant his criteria to be conclusive. The existence of evidence in favor of all criteria does not imply that a causal relation is "proven," nor does the absence of evidence for one or more criteria imply that the relation cannot be causal (see also Rothman & Greenland, 2005). Rather than such a strict interpretation, the criteria were formulated as guidelines to investigate the extent to which the current state of knowledge suggests causation instead of mere association. Furthermore, some criteria (strength, consistency, dose-response curve, and analogy) do not necessarily present strong evidence in favor of causality. However, it is less likely that a causal relation exists when these criteria are not met. Keeping these limitations in mind, we use Hill's criteria as a heuristic framework that guides us in collecting, integrating, and structuring different lines of research. In doing so, the use of the criteria allows us to paint a coherent and comprehensive picture of the current state of the evidence concerning the causal nature of the relation between attentional bias and fear and anxiety.

The Application of Hill's (1965) Criteria on Current Evidence

Strength of the Relation

As a first criterion, Hill (1965) proposed that for a relation between two variables A and B to be causal, there should be firm evidence that A is indeed related to B. In other words, an attentional bias toward threat should be present in high-anxious individuals and not or less so in low-anxious individuals. Because causal relations are often accompanied by strong rather than weak associations, one would expect strong associations between attentional bias and anxiety if the two were causally related. As mentioned earlier, it has been stated or concluded in many recent reviews on cognitive biases in fear and anxiety that there is indeed an association between biased attention and fear and anxiety (e.g., Buckley et al., 2000; Cisler et al., 2009; Cisler & Koster, 2010; Heinrichs & Hofmann, 2001; Mathews & MacLeod, 2005; Mobini & Grant, 2007; Puliafico & Kendall, 2006; Yiend, 2010). However, such narrative reviews cannot give an estimation of the strength of this relation.

The most compelling estimates for the strength of the relation between two variables stem from meta-analytic studies. For this purpose, Bar-Haim et al. (2007) conducted a meta-analysis of 172 studies published up to May 2005 comparing attentional bias in (both clinically and nonclinically) anxious versus nonanxious individuals (see also Table 2). Importantly, their analysis comprised only studies in which performance with threat-related stimuli was compared to performance with neutral stimuli, and in which attentional bias was measured with either a dot probe task, an emotional spatial cueing paradigm, or the emotional Stroop task. Keeping these delineations in mind, Bar-Haim et al. found that anxious individuals showed a larger attentional bias than nonanxious individuals and that this effect was small to moderate in size ($d = 0.41$; Cohen, 1992).² Anxious individuals showed an attentional bias toward threatening stimuli compared to neutral stimuli ($d = 0.45$), whereas nonanxious individuals showed no bias ($d = -0.01$). Although the effect sizes are relatively small, these results demonstrate that there is a statistically reliable association between attentional bias toward threatening stimuli and anxiety.

Consistency

Consistency refers to the extent to which a relation generalizes to different settings. Different aspects of the settings can vary, such as the paradigm that is used to assess attentional bias or the population that is studied. In their meta-analysis, Bar-Haim et al. (2007) included only studies in which attentional bias was assessed with the emotional Stroop task, the dot probe task, or the emotional spatial cueing paradigm. Their analysis revealed that anxious individuals have a larger attentional bias toward threatening stimuli than nonanxious individuals as measured with the emotional Stroop task ($d = 0.45$) and the dot probe task ($d = 0.38$), but not with the exogenous cueing task ($d = 0.01$). Furthermore, they found that individuals suffering from different anxiety problems

² According to Cohen (1992), d values from 0.20 represent small effects, d values from 0.50 represent medium effects, and d values of 0.80 and larger represent large effects.

(including simple phobia, posttraumatic stress disorder (PTSD), panic disorder, generalized anxiety disorder (GAD), and social phobia) have a larger attentional bias than nonanxious controls, with effect sizes ranging between 0.46 and 0.55.

These effect sizes suggest that attentional bias, as measured with different paradigms, is a relatively robust finding across different anxiety problems. However, there is considerable heterogeneity in the effect sizes of individual studies, indicating that the chance of finding attentional bias in a certain sample depends on moderators (e.g., the paradigm that is used, the population, the stimulus materials). For instance, there was no significant difference in attentional bias between high- and low-anxious individuals when attentional bias was assessed with subliminally presented pictures. Unfortunately, Bar-Haim et al. (2007) did not report to what extent this heterogeneity is reduced after controlling for each moderator, which complicates our evaluation of the consistency criterion. In any case, the fact that there is heterogeneity in the effect sizes of individual studies indicates that there are boundary conditions to the finding of a threat-related attentional bias, and thus that there is a certain degree of inconsistency. This heterogeneity remains apparent in more recent research (i.e., studies with the emotional Stroop task, the dot probe task, and the emotional spatial cueing paradigm that were published between April 2005 and May 2011; see Tables 3, 4, and 5).³ In the remainder of this section, we briefly discuss the findings that have been obtained with two paradigms that have not been included in the meta-analysis, namely, the attentional blink task (see also Table 6) and the visual search task (see also Table 7). Considering this evidence allows us to obtain a more general picture of the consistency of the relation between attentional bias and anxiety.

Using the attentional blink task, Arend and Botella (2002) found that emotional T1 words reduced the blink effect in high-trait-anxious individuals but not in low-trait-anxious individuals. However, they did not specify whether all emotional words were threatening or whether some were positive. The results of Lystad, Rokke, and Stout (2009) are in line with those of Arend and Botella. These authors demonstrated a smaller attentional blink with anxiety-related T1 stimuli relative to neutral T1 stimuli when participants were in a negative state. In contrast, Peers and Lawrence (2009) found no effect of state anxiety on the attentional blink following a fearful face distractor. The findings of Barnard, Ramponi, Battye, and Mackintosh (2005) are in direct conflict with those of Arend and Botella and Lystad et al., as they found a *larger* attentional blink after threat-related distractor stimuli in high-state-anxious individuals. To our knowledge, only one study has investigated the effect of the manipulation of the threat value of T2 in a sample of subclinically anxious versus nonanxious individuals. In that study, Fox, Russo, and Georgiou (2005) found that only high-anxious individuals had a reduced attentional blink for fear-relevant T2 stimuli compared to positive T2 stimuli.

Two recent studies by de Jong, Koster, Van Wees, and Martens (2009, 2010) investigated the attentional blink in individuals suffering from social phobia. Both studies provided evidence for the hypothesis that emotional faces are processed more efficiently than neutral faces. However, neither of the studies found a difference in the processing of emotional faces in high versus low socially anxious participants, suggesting that an attentional bias toward socially threatening or relevant stimuli is not specific for high socially anxious individuals.

With respect to more specific fears, three studies have investigated attentional bias toward spiders using the attentional blink paradigm. Cisler, Ries, and Widner (2007) presented spider-related T1 stimuli in both spider fearful and nonfearful participants. They found that the attentional blink was less extended in time in high-fearful compared to low-fearful individuals. However, they did not report on differences between high- and low-fearful individuals in T2 accuracy, which renders their results inconclusive. Reinecke, Rinck, and Becker (2008) claimed that individuals with clinical levels of spider fear preferentially recalled spider-related T2 stimuli compared to nonfearful controls. However, their data should be interpreted with caution because the crucial three-way interaction between anxiety group (spider fearful vs. control), T2 type (spider, snake, neutral or positive), and temporal lag was small ($f = 0.14$; see Cohen, 1992) and not significant ($p = .43$). Hence, their analyses indicated that both fearful individuals and controls showed a reduced attentional blink for spider T2 relative to other T2 types. To our knowledge, only one study (Trippe, Hewig, Heydel, Hecht, & Miltner, 2007) has convincingly demonstrated that the attentional blink effect is reduced for spider-related T2 stimuli in participants with clinical spider fear relative to controls. Finally, Amir, Taylor, Bomyea, and Badour (2009) found that participants suffering from PTSD showed a reduced attentional blink for a neutral T2 stimulus after the presentation of a threatening T1.

In sum, research investigating attentional bias in the context of fear and anxiety using the attentional blink paradigm has produced inconsistent results. Furthermore, it is often unclear how the results of individual studies should be interpreted. Findings of a reduced attentional blink for neutral T2 stimuli after threat-related T1 stimuli in anxious groups have been interpreted as consistent with the notion of an attentional bias toward threat in anxious individuals (Amir, Taylor, et al., 2009; Arend & Botella, 2002). However, Ihssen and Keil (2009; see also Mathewson et al., 2008) found that arousing T1 stimuli *impaired* the identification of neutral T2 stimuli and thus increased the attentional blink. Given the assumption that threat-related words are more arousing for anxious compared to nonanxious individuals, using threatening stimuli as T1 should impair the detection of T2 in anxious groups rather than promote it.

Using the visual search task, researchers have convincingly shown that, relative to control participants, people suffering from clinical GAD exhibit more interference by threatening distractors (Mathews, May, Mogg, & Eysenck, 1990; Mathews, Mogg,

³ These tables contain a summary of all studies relevant to the criterion of consistency that could be found with the following search command on Web of Science: Topic = (attention* bias* AND (fear OR anxiety OR distress OR PTSD OR OCD OR stress OR emotion* OR phobi*) AND ((probe) OR (cue*) OR (stroop))). The results of this search command were further refined for publication year (between April 2005 and May 2011) and language (English). Next, we inspected titles and abstracts and removed literature reviews and articles that were not related to fear or anxiety (e.g., articles about attentional bias in eating disorders or depression). Finally, we only selected studies with an adult sample (for studies with children, see Temporality section), in which single attentional bias scores in anxious individuals were compared to nonanxious controls, and in which the paradigm did not deviate from how we described them earlier in this article. Lists of studies on attentional bias in fear and anxiety that we did not include in the tables based on these last criteria as well as our rationale for not including them are provided in Appendices A, B, and C.

Table 3
Recent Findings Relating Attentional Bias to Fear/Anxiety Using the Emotional Stroop Paradigm

Anxiety problem	Study	N	Evidence ^a	Detail
TA	Edwards et al. (2010b)	64	±	More SI on threat words in high TA group, but results depend strongly on block order and presentation duration
TA	Edwards et al. (2010a)	64	±	More SI on threat words in high TA group, but results depend strongly on block order, presentation duration, and StA manipulations
TA	Reinholdt-Dunne et al. (2009)	56	—	No difference between high and low TA group in SI for threatening words
AS	Teachman (2005)	103	—	No more SI for threatening words in AS group
GA	Sass et al. (2010)	83	—	No clear difference between anxious and nonanxious group in SI for threatening words
HA	Karademas et al. (2008), Experiment 1	51	+	More SI for illness words in HA group
HA	Karademas et al. (2008), Experiment 2	69	+	More SI for illness words in HA group
HA	Witthöft et al. (2008)	107	±	Some evidence for more SI for illness and symptom words in HA group, but results depend strongly on counterbalancing and block numbers
Clinical phobias	Elsesser et al. (2006)	82	+	More SI on phobia-related stimuli in phobic group
SpF	Kwakkenbos et al. (2010)	68	+	More SI on spider stimuli in SpF group
SpF	Olatunji et al. (2008)	57	±	More SI on spider stimuli in SpF group, but only when previously exposed to a spider
Clinical SpF	Kolassa et al. (2005)	38	—	No more SI on spider stimuli in SpF group
Clinical PD	Teachman et al. (2007)	81	+	More SI on panic-related words in PD group
Clinical PD	Lim & Kim (2005)	66	±	More SI for threatening words in PD group, but only if the words were presented subliminally
Clinical PD	Reinecke et al. (2011)	45	±	More SI for both panic and social words in PD group, but only if the words were quickly masked
Clinical PD, mixed anxiety	De Cort et al. (2008)	87	—	No more SI on panic words in PD group, and no more SI on threat words in mixed anxious group
Clinical PD, Clinical OCD	van den Heuvel et al. (2005)	50	±	More SI on panic-related words in PD group, but no more SI for OCD words in OCD group
Clinical OCD	Rao et al. (2010)	100	±	More SI on OCD words in OCD group, but Crucial Group × Word Type interaction is not reported
OCD	Moritz et al. (2008)	46	—	No more SI on OCD words in clinical OCD group
Clinical PTSD	Wittekind et al. (2010)	46	—	No differences between PTSD group and control group in SI for trauma-related words

Note. Our search procedure is described in Footnote 3. AS = anxiety sensitivity; GA = general anxiety; HA = health anxiety; OCD = obsessive-compulsive disorder; PD = panic disorder; PTSD = posttraumatic stress disorder; SI = Stroop interference; SpF = spider fear; StA = state anxiety; TA = trait anxiety.

^a Plus (+) indicates evidence relating attentional bias to fear/anxiety, plus/minus (±) indicates mixed findings, and minus (—) indicates null findings or reversed effects.

Kentish, & Eysenck, 1995; Rinck, Becker, Kellermann, & Roth, 2003). These findings suggest that people with clinical GAD have difficulty disengaging attention from threat. Studies on high-trait-anxious participants are less consistent. For instance, Byrne and Eysenck (1995) found that high-trait-anxious participants were faster to detect angry faces in a happy crowd and slower to detect happy targets in an angry crowd. Derakshan and Koster (2010) also found that high-trait-anxious but not low-trait-anxious individuals were slower to detect a happy face in an angry crowd, suggesting a difficulty disengaging attention from threatening stimuli in anxious individuals. However, such impaired disengagement of attention from angry faces was not observed when a neutral face was presented in an angry crowd. In addition, Derakshan and Koster found that highly anxious participants were slower to detect angry targets in a happy crowd, a finding opposite to that of Byrne and Eysenck. Finally, Notebaert, Crombez, Van Damme, De Houwer, and Theeuwes (2011) found no differences between high- and low-trait-anxious individuals (see also Matsu-moto, 2010).

Within the context of social anxiety, Juth, Lundqvist, Karlsson, and Öhman (2005) found in several experiments that both high and

low socially anxious individuals were faster to detect happy faces in neutral crowds than angry or fearful faces in neutral crowds (see also Rinck et al., 2003). These findings contrast with the results of Gilboa-Schechtman, Foa, and Amir (1999; see also Eastwood et al., 2005), who found that both participants with clinical social phobia and control participants were faster to detect angry faces in neutral crowds than happy faces in neutral crowds. This difference was larger for the individuals with social phobia compared to the control group, suggesting an attentional bias toward angry faces in socially anxious individuals. Finally, Gilboa-Schechtman et al. found that participants suffering from social phobia were more distracted by angry crowds compared to neutral crowds if the target was absent, suggesting that these individuals have difficulty disengaging attention away from threat (see also Baños, Quero, & Botella, 2008; Rinck & Becker, 2005). However, such increased interference was not found with neutral targets in happy versus angry crowds, or with happy targets in neutral versus angry crowds.

Results of studies using a search paradigm in specific phobias (most often spider or snake fear) are more consistent. Rinck, Reinecke, Ellwart, Heuer, and Becker (2005) found that the pres-

Table 4

Recent Findings Relating Attentional Bias to Fear/Anxiety Using the Dot Probe Task

Anxiety problem	Study	N	Evidence ^a	Detail
TA	Koster, Crombez, Verschuere, & De Houwer (2006)	42	+	Larger AB for threat in high TA group
TA	Koster, Verschuere, et al. (2005)	43	+	Larger AB for threat in high TA group
TA	Salemink et al. (2007a)	40	±	Larger AB for threat in high TA group, but only if task was to detect the target; no group difference if task was to identify the target
TA	Eldar et al. (2010)	46	–	No difference between high and low TA group in AB for angry faces
AS	Hunt et al. (2006)	75	+	Larger AB for anxiety symptom words in AS group
SoA	Webb et al. (2010), Experiment 1	28	+	Larger AB for threatening words in SoA group
SoA	Helfinstein et al. (2008)	24	±	Larger AB for angry faces in SoA group, but only when the cue pictures were primed with a neutral word (larger AB in nonanxious control group if cue pictures were primed with a threatening word)
SoA	Ononaiye et al. (2007)	82	±	Larger AB for threatening words in SoA group, but only in one out of 16 possible between-group contrasts
Clinical SoA	Sposari & Rapee (2007)	109	±	AB for faces in SoA group, but regardless of face valence (positive, negative, or neutral)
Clinical SoA	Stevens et al. (2009)	38	±	Larger AB for angry faces in SoA group, but only when cue pictures were presented for 175 ms; no group differences when cue pictures were presented for 600 ms
Clinical SoA	Mueller et al. (2009)	27	–	No difference between SoA group and control group in AB for angry faces
SoA	Pineles & Mineka (2005)	91	–	No difference between high SoA and control group in AB for threatening faces
SoA	Roberts et al. (2010)	63	–	No AB for social threat in SoA group; avoidance of social threat in nonanxious controls
HA	Lees et al. (2005)	48	–	No difference between HA and controls in AB for threatening pictures or words
SpF, SnF	Lipp & Derakshan (2005)	69	±	Larger AB for spiders in SpF group, but no difference in bias for snakes in SnF group versus nonanxious controls
SpF	Mogg & Bradley (2006)	36	±	Larger AB for spiders in SpF group with 200-ms presentation durations, but no group difference with 500-ms presentation duration
Clinical phobias	Elsesser et al. (2006)	77	–	No difference between phobia group and controls in AB for phobia-related stimuli
Clinical PD	Reinecke et al. (2011)	45	±	Larger AB for fearful faces in PD group, but only if stimuli were presented subliminally
Clinical PD	Livermore et al. (2007)	40	–	Attentional avoidance of physical threat words in PD group and AB for physical threat words in nonanxious controls
OCD	Amir, Najmi, et al. (2009)	47	±	AB for OCD stimuli in high OCD group, but only in the first of three blocks (no bias in second and third block or in total of the three blocks)
Clinical OCD	Harkness et al. (2009)	36	–	No difference between OCD group and controls in AB for OCD-related stimuli
Trauma	Lindstrom et al. (2011)	45	±	Larger AB for threat in high-trauma-exposed group, but no difference in trauma symptomatology between the two groups
War exposure (trauma)	Bar-Haim et al. (2010)	131	–	Attentional avoidance of threat in war-exposed group
Trauma	Elsesser et al. (2005)	61	–	No difference between trauma-exposed and control group in AB for trauma-related stimuli

Note. Our search procedure is described in Footnote 3. AB = attentional bias; AS = anxiety sensitivity; HA = health anxiety; OCD = obsessive-compulsive disorder; PD = panic disorder; SnF = snake fear; SoA = social anxiety; SpF = spider fear; TA = trait anxiety.

^a Plus (+) indicates evidence relating attentional bias to fear/anxiety, plus/minus (±) indicates mixed findings, and minus (–) indicates null findings or reversed effects.

ence of spider distractors impaired the detection of targets in participants with clinical spider phobia but not in control participants, suggesting that spider phobia is related to a difficulty disengaging attention away from spiders (see also Gerdes, Alpers, & Pauli, 2008; Lipp & Waters, 2007; Miltner, Krieschel, Hecht, Trippe, & Weiss, 2004, Experiments 1 and 2; but see Öhman et al., 2001; Soares, Esteves, & Flykt, 2009). The study by Rinck et al.

(2005) also offered some evidence for the hypothesis that spider targets more readily attract attention in people with spider phobia (see also Flykt & Caldara, 2006; Miltner et al., 2004, Experiment 2; Öhman et al., 2001; Soares, Esteves, & Flykt, 2009; but see Miltner et al., 2004, Experiment 1), although close inspection of their data suggests that this result is driven mainly by slow responses on spider targets in the control group. Soares, Esteves,

Table 5

Recent Findings Relating Attentional Bias to Fear/Anxiety Using the Emotional Spatial Cueing Paradigm

Anxiety problem	Study	N	Evidence ^a	Detail
TA	X. Li et al. (2005)	30	+	Larger AB for threat in high TA group
TA	Mogg et al. (2008)	42	+	Larger AB for threat in high TA group
TA	Koster, Crombez, Verschuere, Van Damme, & Wiersema (2006)	84	±	Larger AB for threat in high TA group, but only at 100-ms cue presentation duration (avoidance of threat in high TA group with 200-ms and 500-ms cue presentation duration)
TA	Koster, Leyman, et al. (2006)	144	–	No group differences in AB for threat
Clinical OCD	Moritz & von Mühlenen (2008)	55	–	No difference between OCD and control group in AB for OCD-related stimuli

Note. Our search procedure is described in Footnote 3. AB = attentional bias; OCD = obsessive-compulsive disorder; TA = trait anxiety.

^a Plus (+) indicates evidence relating attentional bias to fear/anxiety, plus/minus (±) indicates mixed findings, and minus (–) indicates null findings or reversed effects.

Lundqvist, and Öhman (2009) also found that individuals with elevated levels of spider fear were faster to detect spider targets in neutral crowds compared to snake or neutral targets, whereas nonfearful controls responded equally fast to both spider and snake targets. However, in participants who were afraid of snakes, they did not find faster detection of snake targets than spider targets. Hence, the search pattern for snake fearful participants in this study was more similar to the control group than to the spider fearful group.

In two studies, Pineles and colleagues (Pineles, Shipherd, Mostoufi, Abramovitz, & Yovel, 2009; Pineles, Shipherd, Welch, & Yovel, 2007) found some evidence that trauma-related distractors interfered with target detection in individuals suffering from clinical PTSD, although this result depended in both studies on a

specific task order. Finally, one study reported evidence for attentional bias toward negative faces in clinical panic disorder but not clinical obsessive-compulsive disorder (OCD; Eastwood et al., 2005).

In sum, the meta-analytic findings of Bar-Haim et al. (2007) suggest that a larger attentional bias toward threat in anxious versus nonanxious individuals is a relatively reliable phenomenon across different measures and anxiety disorders. Several studies have convincingly demonstrated a stronger attentional bias for threat in anxious individuals than in controls. Although we endorse the conclusions of Bar-Haim et al., our review of the recent literature uncovered a considerable number of null findings, reversed effects, and (partial) failures to replicate previous results. Crucially, the analysis of Bar-Haim et al. included only published

Table 6

Findings Relating Attentional Bias to Fear/Anxiety Using the Attentional Blink Task

Anxiety problem	Study	N	Evidence ^a	Detail
StA	Lystad et al. (2009)	19	–	Reduced attentional blink for neutral T2 after threatening T1 in high StA group
StA	Peers & Lawrence (2009)	56	–	No difference between high and low StA groups in attentional blink after threatening distractor
TA, StA	Barnard et al. (2005), Experiment 1	48	±	Larger attentional blink after threatening distractor in high StA, but no differences related to high TA, and crucial between-group interaction is not reported
TA	Fox et al. (2005)	28	+	Smaller attentional blink for threatening T2 in high TA group
TA	Arend & Botella (2002)	49	–	Reduced attentional blink for neutral T2 after threatening T1 in high TA group
SoA	de Jong et al. (2009)	67	–	No difference between high SoA and low SoA group in attentional blink after emotional T1
SoA	de Jong et al. (2010)	61	–	No difference between high SoA and low SoA group in attentional blink after emotional T1
Clinical SpF	Trippe et al. (2007)	30	+	Smaller attentional blink for spider-related T2 in SpF group
SpF	Cisler et al. (2007)	80	±	Attentional blink extends over more lags in nonanxious controls, but conventional analyses comparing accuracy of groups per lag are not reported
Clinical SpF	Reinecke et al. (2008)	67	–	No difference between SpF and control group in attentional blink for spider-related T2
PTSD	Amir, Taylor, et al. (2009)	30	–	Reduced attentional blink for neutral T2 after threatening T1 in PTSD group

Note. Our search procedure was identical to the one described in Footnote 3, except we replaced ((probe) OR (cue)) OR (stroop)) by (blink) and retained studies from all years up to May 2011. PTSD = posttraumatic stress disorder; SoA = social anxiety; SpF = spider fear; StA = state anxiety; TA = trait anxiety; T1 = first target; T2 = second target.

^a Plus (+) indicates evidence relating attentional bias to fear/anxiety, plus/minus (±) indicates mixed findings, and minus (–) indicates null findings or reversed effects.

Table 7

Findings Relating Attentional Bias to Fear/Anxiety Using the Visual Search Task

Anxiety problem	Study	N	Evidence ^a	Detail
TA	Byrne & Eysenck (1995)	25	+	Facilitation for threatening targets and more interference by threatening distractors in high TA group
TA	Derakshan & Koster (2010)	77	±	More interference by threatening distractors in high TA group, but only with positive and not with neutral targets; slower detection of threatening target in high TA group
TA	Matsumoto (2010)	42	—	No group differences in interference by threatening distractors or in detection of threatening targets
TA	Notebaert et al. (2011)	45	—	No difference between high and low TA groups
Clinical GAD	Mathews et al. (1990)	36	+	More interference by threatening distractors in GAD group
Clinical GAD	Mathews et al. (1995)	47	+	More interference by threatening distractors in GAD group
Clinical GAD	Rinck et al. (2003), Experiment 2	52	+	More interference by threatening distractors in GAD group
Clinical GAD, clinical SoA	Rinck et al. (2003), Experiment 1	92	±	More interference by threatening distractors in GAD group, but not more interference by speech-related distractors in SoA group
Clinical SoA	Gilboa-Schechtman et al. (1999)	33	±	Facilitation for angry targets with neutral (but not happy) distractors in SoA group; limited evidence for increased interference by angry distractors in SoA group
Clinical SoA	Baños et al. (2008)	65	—	No facilitation for socially threatening targets in SoA group; inconclusive analyses of interference
SoA	Juth et al. (2005), Experiment 1	32	—	No difference between high and low SoA groups in detection of angry targets
SoA	Juth et al. (2005), Experiment 2	32	—	No difference between high and low SoA groups in detection of angry targets
Clinical SoA	Juth et al. (2005), Experiment 3	30	—	No difference between and low SoA groups in detection of angry targets
Clinical SoA	Rinck & Becker (2005)	90	—	Both SoA group and nonanxious controls show more interference by socially threatening distractors compared to neutral distractors; no facilitation for socially threatening targets
Clinical SoA, clinical PD, clinical OCD	Eastwood et al. (2005)	59	±	Facilitation for negative targets with neutral distractors in SoA group and PD group, but not in OCD group
Clinical SpF	Gerdes et al. (2008)	42	+	More interference by spider distractors in SpF group
Clinical SpF	Miltner et al. (2004), Experiment 1	28	+	More interference by spider distractors in SpF group
Clinical SpF	Miltner et al. (2004), Experiment 2	26	+	More interference by spider distractors and facilitation for spider targets in SpF group
Clinical SpF	Rinck et al. (2005), Experiment 1	48	+	More interference by spider distractors in SpF group
Clinical SpF	Rinck et al. (2005), Experiment 2	49	+	More interference by spider distractors in SpF group
SpF, SnF	Lipp & Waters (2007)	23	+	More interference by fear-relevant distractors
SnF/SpF	Soares, Esteves, & Flykt (2009)	32	+	Facilitation for feared targets in SnF/SpF group
SpF, SnF	Flykt & Caldara (2006), Experiment 1	27	±	Facilitation for snake targets in SnF group, but no difference in detection of spider targets between SpF group and nonanxious controls
SnF, SpF	Soares, Esteves, Lundqvist, & Öhman (2009)	60	±	Facilitation for spider targets in SpF group, but no facilitation for snake targets in SnF group
SpF/SnF	Öhman et al. (2001), Experiment 3	34	±	Facilitation for feared target in SnF/SpF group, but not more interference by feared distractors in SnF/SpF group
Clinical PTSD	Pineles et al. (2007)	57	±	More interference by threatening distractors in PTSD group, but results depend on block order; no facilitation for threatening targets in PTSD group
Clinical PTSD	Pineles et al. (2009)	43	±	More interference by trauma-related distractors in PTSD group, but results depend on block order; no facilitation for trauma-related targets in PTSD group

Note. Our search procedure was identical to the one described in Footnote 3, except we replaced ((probe) OR (cue*) OR (stroop)) by (visual search) and retained studies from all years up to May 2011. GAD = generalized anxiety disorder; OCD = obsessive-compulsive disorder; PD = panic disorder; PTSD = posttraumatic stress disorder; SnF = snake fear; SnF/SpF = snake or spider fear; SoA = social anxiety; SpF = spider fear; TA = trait anxiety. ^a Plus (+) indicates evidence relating attentional bias to fear/anxiety, plus/minus (±) indicates mixed findings, and minus (—) indicates null findings or reversed effects.

studies. As such, their estimates are likely to be slight overestimates, vulnerable to publication bias, and the file-drawer problem (Dwan et al., 2008). Indeed, Kimble, Frueh, and Marks (2009) reviewed the literature on emotional Stroop effects in PTSD, comparing findings from peer-reviewed literature and dissertation

abstracts. Whereas 44% of the peer-reviewed articles supported the existence of the effect, this number was significantly smaller in the dissertation abstracts, with only 8% of dissertation abstracts supporting the effect. Kimble et al. therefore concluded that the effect size of the emotional Stroop effect in PTSD is overestimated

because of publication bias, questioning the reliability of the emotional Stroop effect in PTSD.

The belief that attentional bias in fear and anxiety is a reliable, robust, and well-replicated finding may also be due to the strong claims that many authors make about their data in titles, abstracts, or discussion sections. Close inspection of the data often warrants more cautious interpretations because effects are often small, non-significant, or confined to certain blocks, certain presentation durations, or certain subgroups of participants. Table 8 further illustrates the relatively high level of inconsistency. For instance, recent studies using the dot probe task have yielded inconsistent results across different anxiety problems. Only four studies (or 17% of the reviewed studies) provided straightforward evidence for a larger attentional bias toward threat in anxious versus non-anxious individuals, whereas 10 studies (42%) provided mixed or inconclusive findings and another 10 studies (42%) found no group differences or reversed effects. The same holds when considering the level of consistency across different paradigms. For instance, attentional bias in social anxiety is a relatively uncommon finding across different paradigms. These inconsistencies may be caused by differences among the paradigms in their ability to capture certain subcomponents of attention or by the poor psychometric properties of attentional bias measures. Given the poor convergent validity of attentional bias measures, inconsistency across different paradigms is not surprising. It is possible that attentional bias is more consistent than our review suggests but that our current measures of attentional bias lack the appropriate psychometric properties to reveal such a more consistent pattern.

Regardless of their origins, certain inconsistencies do exist. Therefore, we are inclined to adopt a more cautious position than Bar-Haim et al. (2007), who concluded that attentional bias is a well-replicated and consistent phenomenon that, on average, has a

medium effect size. Although we endorse the conclusion that attentional bias is a genuine phenomenon that occurs in different anxiety disorders and can be measured using different paradigms, the existence of multiple null findings and failures to replicate leads us to conclude that the relation between attentional bias and fear and anxiety is probably less consistent than what is generally assumed. Nevertheless, the existence of a certain degree of inconsistency does not exclude the possibility of a causal relation between attentional bias and fear and anxiety. Although inconsistencies make it less likely that attentional bias always and invariably leads to fear and anxiety, it is still possible that attentional bias is a latent vulnerability factor or that attentional bias partially maintains or exacerbates fear and anxiety.

Dose-Response Curve

According to Hill (1965), causal relations are often characterized by a dose-response curve. The most convincing evidence for the existence of a dose-response curve would consist of strong positive correlations between attentional bias to threat and the severity of anxiety problems. Although correlations do not imply causation, many causal relations give rise to significant correlations. However, in the bulk of studies on attentional bias in fear and anxiety, researchers have used factorial designs, using an extreme groups design comparing high- and low-anxious subsamples. Although such studies can show that there is an association between attentional bias and fear and anxiety, they do not indicate whether this association shows a dose-response curve. In order to establish that the individual variations in one variable are related to individual variations in another variable, both variables should have more than two possible values.

Table 8

Consistency Across Paradigms and Anxiety Problems: Number of Experiments Reviewed in the Consistency Section Providing Evidence Relating Attentional Bias to Fear/Anxiety, Providing Mixed or Inconsistent Findings, or Providing Reversed or Null Findings

Anxiety problem	Stroop			Dot probe			Cueing			Attentional blink			Visual search			Consistency across paradigms (%)		
	+	±	–	+	±	–	+	±	–	+	±	–	+	±	–	+	±	–
StA	0	0	0	0	0	0	0	0	0	0	1	2	0	0	0	0	33	67
TA	0	2	1	2	1	1	2	1	1	1	0	2	1	1	2	33	28	39
AS	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	50	0	50
GAD	0	0	1	0	0	0	0	0	0	0	0	0	4	0	0	80	0	20
SoA	0	0	0	1	4	3	0	0	0	0	0	2	1	1	6	11	28	61
HA	2	1	0	0	0	1	0	0	0	0	0	0	0	0	0	50	25	25
SpF, SnF, phobias	2	1	1	0	2	1	0	0	0	1	1	1	7	3	0	50	35	15
PD	2	2	1	0	1	1	0	0	0	0	0	0	1	0	0	38	38	25
OCD	0	1	2	0	1	1	0	0	1	0	0	0	0	0	1	0	29	71
PTSD, trauma	0	0	1	0	1	2	0	0	0	0	0	1	0	2	0	0	43	57
Consistency across anxiety problem (%)	29	33	38	17	42	42	40	20	40	17	17	67	47	23	30	30	29	40

Note. Columns with plus (+) present the number of studies providing evidence relating attentional bias to fear/anxiety (as reported in Tables 3–7), columns with plus/minus (±) reflect the number of studies reporting mixed findings, and columns with minus (–) reflect the number of studies reporting null findings or reversed effects. Numbers in this table deviate slightly from those reported in Tables 3–7 because some experiments fit in more than one cell. AS = anxiety sensitivity; GAD = generalized anxiety disorder; HA = health anxiety; OCD = obsessive-compulsive disorder; PD = panic disorder; PTSD = posttraumatic stress disorder; SnF = snake fear; SoA = social anxiety; SpF = spider fear; StA = state anxiety; TA = trait anxiety.

In several studies, researchers treated fear or anxiety as continuous variables, allowing for the calculation of correlations between attentional bias indices and measures of fear and anxiety (see also Table 9). Most of these studies focused on the association between attentional bias toward threat and state and/or trait anxiety. Overall, the pattern of correlations in these studies is complex. For instance, although Mogg et al. (2000) found that Stroop interference was correlated with both state and trait anxiety, these correlations turned nonsignificant when the authors controlled for social desirability. The latter finding suggests that social desirability is a more important predictor of threat interference than state or trait anxiety. Furthermore, Mogg et al. found that attentional bias toward threat as measured with a dot probe task was not correlated with either state or trait anxiety (see also Fox, Cahill, & Zougkou, 2010; Koster, Leyman, De Raedt, & Crombez, 2006). Next, whereas Bradley, Mogg, and Millar (2000) found a significant correlation between attentional bias toward threat and state but not trait anxiety (see also Dresler et al., 2009; Mogg, Bradley, De Bono, & Painter, 1997), Koster, Crombez, Verschuere, and De Houwer (2004) found that attentional bias toward highly threatening pictures was correlated with trait but not state anxiety. Salemink et al. (2007a) found a modest positive correlation between attentional bias and trait anxiety, but only if attentional bias was

measured in a dot probe task in which participants were instructed to respond to the location of the target (above or below fixation), and not when participants had to respond to the identity of different targets. Finally, Applehans and Luecken (2006) even reported a negative correlation between attentional bias toward socially threatening words in a dot probe task and trait anxiety (for a similar negative correlation, see Putman, 2011).

The same inconsistencies hold for the correlation between attentional bias indices and more specific fears. For instance, Lipp and Derakshan (2005) found that attentional bias toward pictures of spiders correlated significantly with participants' self-reported level of spider fear. However, they did not find a significant correlation between attentional bias for snakes and participants' levels of snake fear. Furthermore, Van Bockstaele et al. (2011a) did not find a significant correlation between attentional bias toward spiders and self-reported spider fear. This brief summary suggests that the correlation between attentional bias on the one hand and fear and anxiety on the other is fairly weak and not robust.

Another way to address the dose-response curve is through studies in which three or more groups of participants differing in fear or anxiety level are included. Unfortunately, such studies are rare. In one study, Bradley et al. (2000) split their sample in

Table 9
Summary of Evidence for a Dose-Response Curve

Anxiety problem	Study	N	AB measures	Evidence ^a	Detail
StA, TA	Dresler et al. (2009)	50	Stroop	±	Positive association between SI and StA but not TA
StA, TA	Koster, Crombez, Verschuere, & De Houwer (2004)	44	Dot probe	±	Positive correlation between AB for highly threatening pictures and TA, but no significant correlations with AB for mildly threatening pictures or StA
StA, TA	Mogg et al. (1997)	33	Dot probe	±	Positive correlation between AB for threat and StA, but no significant correlation between AB for threat and TA
StA, TA	Mogg et al. (2000)	60	Stroop and dot probe	±	Positive correlation between SI and both TA and StA, but no correlations between attentional bias in dot probe task and either trait or state anxiety
StA, TA	Salemink et al. (2007a)	133	Dot probe (two versions)	±	Positive correlation between AB for threat and TA, but only in one version of the dot probe, and no significant correlations between AB and StA
StA, TA	Fox et al. (2010)	104	Dot probe	–	No significant correlations between AB for negative pictures and TA or StA
TA	Applehans & Luecken (2006)	63	Dot probe	–	Negative correlation between AB for threat and TA
TA	Putman (2011)	40	Dot probe	–	Negative correlation between AB and TA
GA, stress	Koster, Leyman, et al. (2006)	144	Exogenous cueing	–	No significant correlations between AB for negative cues and GA or stress
SoA, StA, TA	Bradley et al. (2000)	54	Dot probe	±	Positive correlation between AB for threat and StA, but not TA or SoA
SnF, SpF	Lipp & Derakshan (2005)	69	Dot probe	±	Positive correlation between AB for spiders and SpF, but no significant correlation between AB for snakes and SnF
SpF	Van Bockstaele et al. (2011a)	53	Dot probe	–	No significant correlation between AB for spiders and SpF

Note. For these studies, we used the following search command on Web of Science: Topic = (attention* bias* AND (fear OR anxiety OR distress OR PTSD OR OCD OR stress OR emotion* OR phobi*)) AND ((probe) OR (cue*) OR (stroop) OR (blink) OR (search)). The results of this search command were further refined for language (English) and publication year (before May 2011), and we only selected original research articles. Next, we inspected titles and abstracts and removed articles that were not related to fear or anxiety. Finally, we selected the studies in which anxiety was treated as a continuous variable and in which correlations were reported. AB = attentional bias; GA = general anxiety; SI = Stroop interference; SnF = snake fear; SoA = social anxiety; SpF = spider fear; StA = state anxiety; TA = trait anxiety.

^a Plus/minus (±) indicates mixed findings of correlations, and minus (–) indicates null findings or negative correlations. Note that the + score, reflecting positive correlations and thus indicating that anxiety continuously increased with increased attentional bias, is not applicable in this table.

three groups based on their participants' anxiety scores. They found that both medium- and high-state-anxious participants showed an attentional bias toward threatening faces, without differences between these two groups. When they split their sample based on participants' trait anxiety scores, they did not observe differences in attentional bias between the low-, medium-, and high-anxious group. In line with these results, Bar-Haim et al. (2007) found no difference in attentional bias between participants with high self-reported anxiety and clinically anxious participants. This latter result confirms that higher levels of fear and anxiety are not necessarily linked to higher levels of attentional bias.

In sum, the evidence for a dose-response curve in the relation between attentional bias toward threat and fear and anxiety is limited and often contradictory. At least three methodological factors may in part account for the relatively weak evidence for a dose-response curve. First, the significance levels for the correlations in the studies that we discussed above were never corrected for multiple statistical tests. This may have resulted in a number of false positive results, hence distorting the overall pattern of findings. Related to this issue, correlation coefficients also give an estimate of the size of an effect, regardless of their significance. According to Cohen (1992), correlations of .10 reflect small effects, correlations around .30 reflect medium effects, and correlations of .50 and larger reflect large effects. In the studies that we discussed, the correlations were usually between .20 and .40, suggesting a medium-sized effect. However, this may be an overestimation because authors often do not report the exact values of nonsignificant correlations. To estimate the average size of the correlation between attentional bias and measures of fear or anxiety, authors should report all the relevant correlations, not only the significant ones.

Second, the statistics that are commonly used to assess a dose-response relation rely on the assumption that the relation between two variables is monotonic and linear. Hence, in order to detect significant correlations between attentional bias and fear or anxiety, the relation between attentional bias and fear and anxiety must be monotonic and linear. Some authors have indeed proposed a linear relation, so that the stronger the attentional bias is, the more anxious the individual is (e.g., Williams et al., 1997). However, according to the cognitive-motivational view of Mogg and Bradley (1998), both high- and low-anxious individuals can show an attentional bias for threat, depending on the threat value of the stimulus. According to this view, both high- and low-anxious individuals will attend to highly threatening stimuli, but only high-anxious people will attend to mildly threatening stimuli. If this is indeed the case, variations in the correlations between attentional bias and anxiety measures may be caused by the varying levels of threat of the stimuli that are used in different studies.

Third, the poor psychometric properties of the attentional bias measures, and their poor reliability in particular, are a likely cause of the divergence in findings. The poor reliability of measures of attentional bias implies that the measures capture a lot of noise. As a result, the measures will not correlate strongly with levels of fear or anxiety, even if attentional bias is causally related to fear and anxiety.

Plausibility

The criterion of plausibility requires the assumed causal mechanism to make sense in the light of current knowledge and theories. As we have argued earlier, several models or theories assume that attentional bias may be a cause of fear and anxiety. For instance, the theory of Eysenck (1992) specifies a plausible mechanism through which attentional bias can causally influence fear and anxiety. He argued that hypervigilance for threatening stimuli increases the likelihood that such stimuli are perceived in the environment. As a result, hypervigilant individuals are thought to be more likely to perceive their environment as threatening, which will in turn increase the individual's anxiety level. Unfortunately, to our knowledge, there are no empirical data in which this hypothesis is tested.

There are also theories that make the idea that attentional bias causally influences fear and anxiety less plausible. Several theories of classical (fear) conditioning consider attention a prerequisite for learning (N. J. Mackintosh, 1975; Mitchell, De Houwer, & Lovibond, 2009; Wagner, 1981). In a prototypical fear conditioning paradigm, one of two previously neutral cues (the reinforced conditional stimulus [CS+]) is paired with an aversive stimulus (unconditioned stimulus [US]), whereas a second neutral stimulus (nonreinforced conditional stimulus [CS-]) is not paired with the US. Because of the pairings with the US, the CS+ typically gains a higher threat value than the CS-. Dawson (1970) found that learning in such a fear-conditioning paradigm was impaired when participants were engaged in an attention-consuming secondary task. Importantly, the extinction of a conditioned fear response (i.e., the reduction of fear following the presentation of the CS+ in absence of the US) is considered to be the result of the learning of a new CS+/no-US relationship (Bouton, 2002). Given the idea that learning requires attention, the strength of extinction should also depend upon the availability of attentional resources. In other words, attending to signals of threat is necessary to successfully learn the CS+/no-US relation. Thus, attending to threat should reduce fear rather than promote it.

A similar line of reasoning is also provided by the emotional processing theory of exposure therapy (Foa & Kozak, 1986). According to this theory, fear is the result of the activation of a fear structure. To reduce fear, the individual should be exposed to information that is incompatible with the information that is stored in the fear structure. For instance, an individual suffering from spider phobia may believe that spiders are aggressive and poisonous, and thus dangerous. During exposure therapy, the individual will be confronted with spiders in absence of such negative consequences, and as a result such negative beliefs will be negated. Importantly, the fear structure can only be changed when the individual attends to the threatening stimulus. Hence, attending to threat should reduce fear rather than promote it. However, contrary to the theory, it should be noted that the influence of attention versus distraction during exposure is still a topic of debate (e.g., Johnstone & Page, 2004; Schmid-Leuz, Elsesser, Lohrmann, Jöhren, & Sartory, 2007).

Temporality

To meet the criterion of temporality, the data must show that attentional bias toward threat precedes the onset of fear or anxiety. One way of investigating temporality is through the

induction of attentional bias toward threat in nonanxious participants who do not show a bias at baseline. If such an experimentally induced attentional bias toward threat results in increased levels of fear or anxiety, this would indeed support the idea that attentional bias precedes the onset of anxiety in time. However, because this type of research involves an experimental manipulation, we discuss these studies in detail in the section on experimental evidence.

In addition to experimental studies, temporality can be examined through prospective, longitudinal designs. In prospective studies, researchers have measured attentional bias at Time 1 to predict fear or anxiety responses at Time 2 while controlling for participants' levels of fear or anxiety at Time 1 (see Table 10). If attentional bias precedes fear and anxiety in time, then higher levels of attentional bias at Time 1 should be associated with elevated anxiety at Time 2. In a seminal study,

MacLeod and Hagan (1992) administered an emotional Stroop task to women who were awaiting a medical examination. They found that the amount of Stroop interference was a significant predictor of the intensity of the self-reported emotional distress that was elicited by a negative medical diagnosis. However, the participants in the study of MacLeod and Hagan were already under stress when their attentional bias was measured. Hence, it is possible that their attentional bias scores were already influenced by their relatively high level of state anxiety. Van den Hout, Tenney, Huygens, Merckelbach, and Kindt (1995) later replicated the findings of MacLeod and Hagan in a sample of currently nonstressed healthy participants, showing again that the amount of emotional Stroop interference was a unique predictor self-rated vulnerability to life stress (see also Nay, Thorpe, Roberson-Nay, Hecker, & Sigmon, 2004; Pury, 2002; Verhaak, Smeenk, van Minnen, & Kraaimaat, 2004).

Table 10

Summary of Evidence for Temporality: Attentional Bias as a Predictor of Fear and Anxiety

Anxiety problem	Study	N	AB measures	Outcome variables	Evidence ^a	Detail
SR	Egloff et al. (2002)	97	Dot probe	Cardiovascular response to social stress	+	AB for socially threatening words predicts cardiovascular reactivity in response to social stressor
SR	Fox et al. (2010)	82/70	Dot probe	Physiological and self-reported response to stress	±	AB for negative pictures predicts physiological response to lab-induced ($N = 82$) and real-life ($N = 70$) stress, but avoidance of negative pictures predicts self-reported anxiety in response to lab-induced stress
SR	MacLeod & Hagan (1992)	15	Stroop	Response to medical exam	+	SI predicts intensity of emotional distress
SR	Nay et al. (2004)	87	Stroop	Response to challenge	+	SI predicts emotional response to challenge
SR	Pury (2002)	30	Stroop	Response to exams	+	SI predicts anxiety in response to upcoming exams
SR	van den Hout et al. (1995)	34	Stroop	Response to stress	+	SI predicts emotional vulnerability to stress
SR	Verhaak et al. (2004)	47	Stroop	Response to medical exam	+	SI predicts change in anxiety following medical outcome
SR	Jansson & Najström (2009)	42	Stroop	Physiological and self-reported response to stress	±	SI predicts self-reported stress, but inverse relation for the prediction of physiological stress
SoA	Pérez-Edgar et al. (2010)	126	Dot probe	Social withdrawal	+	AB for threat moderates the relation between early childhood behavioral inhibition and social withdrawal in adolescence
SoA	Pérez-Edgar et al. (2011)	187	Dot Probe	Social withdrawal	+	AB for threat moderates the relation between early childhood behavioral inhibition and social withdrawal in later childhood
SpF	Van Bockstaele et al. (2011a)	50/48	Dot probe, disengagement task	Behavioral and physiological measures of spider fear	±	AB for spiders predicts changes in heart rate in response to a spider ($N = 50$) but not changes in skin conductance ($N = 48$) or behavioral avoidance of a spider ($N = 50$)

Note. For these studies, we used the following search command on Web of Science: Topic = (attention* bias* AND (fear OR anxiety OR distress OR PTSD OR OCD OR stress OR emotion* OR phobi*)) AND ((probe) OR (cue*) OR (stroop) OR (blink) OR (search)) AND ((predict*) OR (regress*))). The results of this search command were further refined for language (English) and publication year (before May 2011), and we only selected original research articles. Next, we inspected titles and abstracts and removed articles that were not related to fear or anxiety. As a backup, we compared our list of results with all studies citing MacLeod and Hagan (1992), which is, to our knowledge, the earliest article in which attentional bias is used to predict anxiety. Finally, we only selected studies in which individual scores on a measure of attentional bias were used to predict scores on measures of fear or anxiety. AB = attentional bias; SI = Stroop interference; SoA = social anxiety; SpF = spider fear; SR = stress responsiveness.

^a Plus (+) indicates evidence that measures of AB can predict fear and anxiety, whereas plus/minus (±) indicates mixed findings. Note that the minus (−) score, reflecting that measures of AB are clearly unable to predict later fear or anxiety, is not applicable in this table.

Attentional bias is also predictive of physiological measures of fear and anxiety. For instance, Egloff, Wilhelm, Neubauer, Mauss, and Gross (2002) measured attentional bias toward social threat words in a dot probe paradigm. They showed that the attentional bias score was a better predictor of subsequent cardiovascular reactivity in response to a social stressor than scores on the trait version of the State and Trait Anxiety Inventory (Spielberger, Gorusch, Lushene, Vagg, & Jacobs, 1983). Similar results have been reported by Fox et al. (2010), who showed that attentional bias toward negative pictures was predictive of cortisol responses to both laboratory and real-life stressors. Recently, Van Bockstaele et al. (2011a) used measures of attentional bias for spiders and implicit spider attitudes (Implicit Association Test; Greenwald, McGhee, & Schwartz, 1998) to predict changes in heart rate and skin conductance in response to spider-related stimuli and overt approach–avoidance behavior of a spider. They found that attentional bias and more negative implicit spider attitudes were predictive of increases in heart rate, but not of changes in skin conductance or behavioral avoidance. At this point, we are aware of only one study that has reported discrepant findings. In that study, Jansson and Najström (2009) found that higher levels of emotional Stroop interference were predictive of *smaller* increases in skin conductance in response to a stressor. Yet, most published prospective studies showed that attentional bias can predict emotional reactivity to stress and levels of anxiety.

Critical support for the temporality argument could come from developmental studies examining whether childhood attentional bias predicts adult pathological fear or anxiety. There is evidence for the existence of an attentional bias toward threat in early childhood. For instance, LoBue and DeLoache (2008) demonstrated that 3- to 5-year-old children were faster to detect snake targets than neutral targets in a visual search task (see also Dalglish et al., 2003; Waters & Lipp, 2008a, 2008b; for recent reviews on attentional biases in children and adolescents, see Field & Lester, 2010; Puliafico & Kendall, 2006). In their meta-analysis, Bar-Haim et al. (2007) confirmed that fearful and anxious children have a larger attentional bias toward threatening stimuli than nonanxious children. However, the finding of an attentional bias in early childhood does not necessarily mean that attentional bias precedes the onset of fear or anxiety.

A recent line of research provides more convincing evidence for temporality. Pérez-Edgar et al. (2010) first assessed behavioral inhibition in early childhood. Behavioral inhibition is a temperament trait characterized by signs of distress to unfamiliar or novel stimuli (Kagan, Reznick, & Snidman, 1987; L. A. Schmidt et al., 1997), and a risk factor for the development of anxiety disorders at a later age (Hirshfeld et al., 1992). When the participants reached adolescence, Pérez-Edgar et al. measured attentional bias toward angry faces and social withdrawal, which they argued is a precursor or vulnerability factor for elevated trait and social anxiety. They found that the relation between early childhood behavioral inhibition and social withdrawal during adolescence was moderated by attentional bias toward threat. That is, adolescents' childhood levels of behavioral inhibition were only related to their level of social withdrawal at adolescence if they had a relatively large attentional bias toward threat (see also Pérez-Edgar et al., 2011). These results suggest that an attentional bias toward threat is a cognitive vulnerability factor for the development of anxiety problems.

Finally, there is some evidence for the alternative possibility that fear and anxiety may precede the onset of an attentional bias. Several studies have shown that following a fear-conditioning procedure, participants show an attentional bias toward the CS+ relative to the CS− (e.g., Koster, Crombez, Van Damme, Verschuere, & De Houwer, 2004, 2005; Van Damme, Crombez, Eccleston, & Goubert, 2004; Van Damme, Lorenz, et al., 2004). These results suggest that attentional bias originates either simultaneously with or only after the onset of the fear. Regardless of which of these two options is correct, one can conclude that attentional bias does not necessarily precede fear or anxiety. Hence, the results of these fear-conditioning studies allow for the conclusion that attentional bias does not necessarily precede the onset of fear and anxiety.

Besides these fear-conditioning studies, there is also some evidence for the hypothesis that increases in state anxiety are followed by an increase in attentional bias toward threat. For instance, Edwards, Burt, and Lipp (2006; but see Cooper et al., 2011; Edwards, Burt, & Lipp, 2010b) measured Stroop interference in high- and low-trait-anxious individuals, while manipulating state anxiety through exposure to electric shocks. They found that high-trait-anxious participants showed more interference for unmasked threat words than low-trait-anxious participants, but only when state anxiety was elevated. This finding suggests that increased state anxiety is followed by increased attentional bias in high-anxious individuals, and thus that the onset of state anxiety may precede the onset attentional bias toward threat.

In sum, a number of prospective studies suggest that attentional bias toward threat can predict later stress reactivity. Furthermore, there is some preliminary evidence showing that attentional bias toward threat moderates the relation between markers of anxiety in early childhood and adolescence, suggesting that attentional bias toward threat is a cognitive vulnerability factor for the development of fear and anxiety. However, the accurate prediction of fear or anxiety based on measures of attentional bias requires the accurate measurement of an individual's attentional bias. The low reliability of the attentional bias measures is especially troublesome for prediction studies. The measurement of attentional bias in children may even be more unreliable, because some cognitive functions relevant to the measurement of attentional bias are not fully developed until adulthood (Luna, Garver, Urban, Lazar, & Sweeney, 2004; Puliafico & Kendall, 2006). Although Pérez-Edgar et al. (2011) argued that they were able to reliably measure attentional bias in 5-year-old children, they only reported the children's level of response accuracy to support this claim. Clearly, the extent to which children are able to comply with certain instructions and perform a certain task does not reflect the reliability of a measurement. To date, there is no research that assessed the split-half or test–retest reliability of attentional bias measures in children.

These psychometric issues aside, several fear-conditioning studies have shown that attentional bias can also be a consequence of negative learning experiences (B precedes A), illustrating that it is unlikely that attentional bias and fear and anxiety are causally related in a unidirectional manner. However, as several of the prospective studies controlled for baseline levels of anxiety, the existence of a reversed relation between fear and attentional bias (B precedes A) does not negate the finding that attentional bias predicts anxiety and fear (A precedes B). Finally, the most con-

vincing evidence for temporality would be offered by a large-scale longitudinal study in which attentional bias of a nonanxious sample is used to predict lifetime prevalence of fear or anxiety disorders. Unfortunately, there is no such research to date.

Experiments

Experimental studies offer the most compelling evidence for causality. If one can establish that an experimentally induced change in attentional bias toward threatening stimuli is followed by a corresponding change in fear or anxiety, this provides strong evidence for a causal relation between attentional bias and fear or anxiety. However, as mentioned before, even experimental designs are subject to a number of important limitations or biases when not properly performed (Altman et al., 2001). For instance, it is imperative that proper control groups are included and that participants are randomly assigned to experimental and control groups. Also, differences in attrition rates and possible confounding variables should be measured and/or controlled for. In all the studies that we discuss below, participants were randomly assigned to experimental and control groups, and none of the authors reported significant group differences in attrition rates.

With these limitations kept in mind, several studies have recently been published in which researchers investigated the effects of changes in attentional bias on measures of fear and anxiety (for recent reviews, see Bar-Haim, 2010; Browning, Holmes, & Harmer, 2010). In these studies, a standard attentional bias assessment paradigm, such as the dot probe task, is modified to an attentional training task. Most often, this is achieved by changing the proportions of congruent and incongruent trials (for an exception, see Johnson, 2009). Typically, one group of participants is exposed to a high percentage of incongruent trials and is thus trained to avoid threat. Changes in fear and anxiety over time (pre- vs. posttraining) in this “avoid threat group” are compared to changes in fear and anxiety in a group that is trained to direct attention toward the threatening information (“attend threat group”; more congruent trials than incongruent trials) or a no-training control group (equal number of congruent and incongruent trials). Because of the relative importance of experimental evidence for the demonstration of causation, we discuss every experimental study that was published before June 2011 and that addresses the effect of experimentally induced changes in attentional bias on measures of fear and anxiety below (see also Table 11, in which we included a brief description of any additional studies published before April 2013).

MacLeod, Rutherford, Campbell, Ebsworthy, and Holker (2002) were among the first to adopt an attentional bias modification procedure. In two experiments, they trained one group of participants to attend to negative words by presenting them a higher percentage of congruent dot probe trials (i.e., target replaced the negative cue word). A second group was trained to avoid negative words by presenting them a higher percentage of incongruent dot probe trials (i.e., target replaced the positive cue word). After this training phase, all participants were subjected to a stressful anagram task. Crucially, participants in the avoid negative group reported less anxiety and less feelings of depression in response to this stress task than participants in the attend negative group. This finding suggests that changes in attentional bias affected participants’ vulnerability to experience anxiety.

Similar findings have been reported by Dandeneau, Baldwin, Baccus, Sakellaropoulou, and Pruessner (2007), who used an adaptation of a visual search task. In one of their experiments, these authors showed that students trained to find a positive face in a grid of frowning faces reported less stress for an upcoming exam than participants trained to find a particular flower in a grid of other flowers. In a follow-up experiment, they used the same training procedure on a sample of telemarketers. Participants who were trained to attend to positive faces reported less stress, were rated as more self-confident, had lower cortisol levels, and had better sales performances than participants in the flower control group. See, MacLeod, and Bridle (2009) showed that students trained to avoid emotionally negative words in a dot probe task reported smaller increases in anxiety in response to a real life stressor (moving to another country) than participants in a no-training control group. Similar findings have also been reported by Johnson (2009), who (instead of altering the proportions of congruent and incongruent trials) instructed participants in an avoid threat group to attend to smiling faces and avoid angry faces. He found that the avoid threat group reported less frustration in response to a stress task than participants in a control group, who were given no explicit instructions. Finally, Eldar, Ricon, and Bar-Haim (2008) closely replicated the findings of MacLeod et al. (2002) in a sample of 7- to 12-year-old children, demonstrating a smaller increase in self-reported anxiety in response to a stressor in a group of children that was trained to avoid threat compared to an attend threat group.

The only discrepant finding with regard to the effect of attentional training on stress reactivity has been reported by Behar, McHugh, Peckham, and Otto (2010). The main aim of this study was to investigate whether the attentional training manipulation could be strengthened through the administration of D-cycloserine. D-cycloserine is a partial agonist at the N-methyl-D-aspartate receptor in the amygdala that has been shown to increase extinction learning, supposedly by facilitating memory consolidation (e.g., Davis, Myers, Ressler, & Rothbaum, 2005). Behar et al. gave one group of participants D-cycloserine, whereas another group was administered a placebo. Next, all participants completed an attentional training phase in which they were trained to avoid threatening words. After this training phase, participants were subjected to various stress-inducing tasks. To assess distress tolerance, Behar et al. measured how long participants persisted in these tasks, as well as participants’ self-reported feelings of negative affect and anxiety during the tasks. They found that D-cycloserine indeed strengthened the attentional training, as indicated by a stronger decrease in attentional bias in the experimental group compared to the control group. However, they found no group differences in their measures of distress tolerance. This study indicates that the attentional avoidance training effect is not necessarily associated with reduced stress reactivity or anxiety, although the absence of a no-training control group is a notable limitation. Because both groups were trained to avoid threat and both groups showed a reduction in attentional bias, it remains possible that the induced attentional avoidance resulted in a similar reduction in stress vulnerability in both groups.

With regard to GAD and high trait anxiety, five studies to date have investigated the effects of attentional bias modification on anxiety. In a first article, Mathews and MacLeod (2002) summarized two experiments in which they assigned high-trait-anxious

Table 11
Summary of Experimental Evidence: Effects of Changes in Attentional Bias on Fear and Anxiety

Anxiety problem	Study	N	Groups	Training sessions	Trials (total)	Training task	Evidence ^a	Detail
SR	Dandeneau et al. (2007), Experiment 3a	25	Avoid threat, neutral task	5	400	Visual search	+	Less self-reported stress for upcoming exam in avoid threat group
SR	Dandeneau et al. (2007), Experiment 3b	23	Avoid threat, neutral task	5	400	Visual search	+	Less self-reported stress, more observer-rated confidence, and better work performance in avoid threat group
SR	MacLeod et al. (2002), Experiment 1	64	Avoid negative, attend negative	1	576	Dot probe	+	Smaller increase in self-reported negative mood in response to stress task in avoid negative group
SR	MacLeod et al. (2002), Experiment 2	62	Avoid negative, attend negative	1	576	Dot probe	+	Smaller increase in self-reported negative mood in response to stress task in avoid negative group
SR	See et al. (2009)	40	Avoid negative, control	15	2,880	Dot probe	+	Smaller increase in self-reported StA in response to a real-life stressor in avoid negative group
SR	Bar-Haim et al. (2011)	34	Avoid threat, control	2	768	Exogenous cueing	+	Increased self-reported StA in response to stress task in control group; no change in StA in avoid threat group
SR	Eldar et al. (2008)	26	Avoid threat, attend threat	2	672	Dot probe	+	Smaller increase in self-reported StA in response to stress task in avoid threat group
SR	Baert et al. (2012)	28	Avoid threat, control	6	1,260	Dot probe	±	Better physiological stress recovery after interview in avoid threat group, but no group differences during interview
SR	Johnson (2009)	109	Avoid threat, control	1	144	Dot probe	±	Smaller increase in self-reported frustration in response to a stress task in avoid threat group, but no group differences in self-reported anxiety
SR	Behar et al. (2010)	34	“Strong” avoid threat, “weak” avoid threat	1	672	Dot probe	–	No group differences in behavioral stress tolerance or self-reported negative affect during stress task
TA	Eldar et al. (2012)	30	Avoid threat, control	4	1,920	Dot probe	±	Less anxiety symptoms and reduced observer-rated symptom severity in avoid threat group; no group differences in self-reported anxiety
TA	Mathews & MacLeod (2002), Experiment 7	29	Avoid threat, control	10	7,500	Dot probe	±	Stronger decrease in self-reported TA in avoid threat group, but control group had lower TA scores prior to the training; very concise description of methods and results (e.g., no means, no standard deviations)
TA	Mathews & MacLeod (2002), Experiment 8	30	Avoid threat, control	8	6,000	Dot probe	±	Decrease in self-reported TA in avoid threat group and not in control group, but very concise description of methods and results (e.g., no means, no standard deviations, crucial interactions not reported)

(table continues)

Table 11 (continued)

Anxiety problem	Study	N	Groups	Training sessions	Trials (total)	Training task	Evidence ^a	Detail
Clinical GAD	Amir, Beard, Burns, & Bomyea (2009)	29	Avoid threat, control	8	1,280	Dot probe	+	Larger decreases in anxiety measured through both self-report and interviewers in avoid threat group
GAD/worry	Hazen et al. (2009)	24	Avoid threat, control	5	1,080	Dot probe	+	Larger decrease in self-reported symptoms of generalized anxiety in avoid threat group
GAD/worry	Hayes, Hirsch, & Mathews (2010)	48	Avoid threat, control	1	480	Dot probe	±	Less self-reported and observer-rated negative thought intrusions in avoid threat group, but no group differences in self-reported anxiety ratings
Worry	Krebs et al. (2010)	64	Avoid threat, attend threat	1	576	Dot probe	±	Less negative thought intrusions in avoid threat group, but only as rated by an observer in participants who received information about the training contingencies; no group differences in uninformed participants, on self-rated negative thought intrusions, or on negative mood
SoA	Amir et al. (2008)	94	Avoid threat, control	1	128	Dot probe	+	Smaller increase in self-reported anxiety in response to speech task and better speech performance in avoid threat group
Clinical SoA	Amir, Beard, Taylor, et al. (2009)	44	Avoid threat, control	8	1,024	Dot probe	+	Larger decreases in self-reported and observer-rated SoA in avoid threat group
Clinical SoA	N. B. Schmidt et al. (2009)	36	Avoid threat, control	8	1,024	Dot probe	+	Decrease in self-reported and observer-rated SoA in avoid threat group
Clinical SoA	Heeren et al. (2011)	79	Avoid threat, attend neutral, avoid threat and attend neutral, control	1	560	Dot probe or exogenous cueing	±	Less observer-rated anxiety during speech task in avoid threat groups, but no group differences in self-reported mood or anxiety
Clinical SoA	Heeren et al. (2012)	57	Avoid threat, attend threat, control	4	2,976	Dot probe	±	Less physiological stress and better speech performance in avoid threat group; less self-reported SoA in avoid threat group at 2-week follow-up but not immediately after the training
SoA	Higgins & Hughes (2012)	64	Avoid negative, attend negative	1	288	Dot probe	±	More physiological stress in response to speech task in attend negative group, but only in participants high in neuroticism; reversed pattern of results in participants with low neuroticism scores, but no group differences in self-reported anxiety
SoA	S. Li et al. (2008)	24	Avoid threat, control	7	3,360	Dot probe	±	Decrease in self-reported SoA in avoid threat group, but only on one out of three SoA questionnaires
Clinical SoA	Boettcher et al. (2012)	64	Avoid threat, control	8	1,024	Dot probe	–	No group differences in self-reported and observer-rated SoA

Table 11 (continued)

Anxiety problem	Study	N	Groups	Training sessions	Trials (total)	Training task	Evidence ^a	Detail
Clinical SoA	Bunnell et al. (2013)	31	Avoid threat, control	8	1,024	Dot probe	–	No group differences in self-reported and observer-rated SoA, nor in performance during speech tasks
Clinical SoA	Carlbbring et al. (2012)	79	Avoid threat, control	8	1,024	Dot Probe	–	No group differences in self-reported and observer-rated SoA
SoA	Julian et al. (2012)	112	Avoid threat, control	1	128	Dot probe	–	No group differences in self-reported StA in response to speech task
SoA	Klumpp & Amir (2010)	87	Avoid threat, attend threat, control	1	128	Dot probe	–	Smaller increase in self-reported StA in response to a speech task in avoid and attend threat group
Clinical SoA	Neubauer et al. (2013)	48	Avoid threat, control	8	1,024	Dot probe	–	No group differences in self-reported SoA, nor in performance during speech tasks
SpF	Harris & Menzies (1998)	44	Avoid spiders, attend spiders	1	40	Dot probe	–	No group differences in self-reported SpF
SpF	Reese et al. (2010)	41	Avoid spiders, control	1	768	Dot probe	–	No group differences in self-reported SpF, avoidance in a BAT, or self-reported fear during a BAT
SpF	Van Bockstaele et al. (2011b)	65	Avoid spiders, attend spiders	1	288	Dot probe	–	No group differences in self-reported SpF, implicit spider associations, physiological fear responses, or self-reported fear and avoidance in a BAT
Conditioned fear	Van Bockstaele et al. (2010)	67	Avoid threat, attend threat, control	1	72	Exogenous cueing task	–	Faster self-reported extinction of fear for signal of threat in attend threat group
OCD	Najmi & Amir (2010)	52	Avoid threat, control	1	288	Dot probe	±	Avoid threat group completed more steps on a BAT, but no group differences in self-reported anxiety during the BAT or on OCD symptoms questionnaire
Trauma	Verwoerd et al. (2012)	45	Avoid trauma, control	1	384	Exogenous cueing	+	Less self-reported trauma-related memory intrusions in avoid trauma group
PTSD	Schoorl et al. (2013)	76	Avoid trauma, control	8	1,280	Dot probe	–	No group differences in self-reported and observer-rated symptoms of PTSD

Note. For these studies, we used the following search command on Web of Science: Topic = (attention* bias* AND (fear OR anxiety OR distress OR PTSD OR OCD OR stress OR emotion* OR phobi*)) AND ((train*) OR (manipulate*) OR (modification) OR (ABM)). The results of this search command were further refined for language (English) and publication year (before April 2013), and we only selected original research articles. Next, we inspected titles and abstracts and removed articles that were not related to fear or anxiety. As a backup, we compared our list of results with all studies citing MacLeod et al. (2002); Amir, Beard, Burns, and Bomyea (2009); Harris and Menzies (1998); or Amir et al. (2008), which are the earliest articles in which attentional bias is manipulated in order to induce changes in anxiety. Finally, we only selected studies in which attentional bias toward threatening stimuli was manipulated in one group and changes in fear or anxiety were compared with a control group. BAT = behavioral approach task; GAD = generalized anxiety disorder; OCD = obsessive-compulsive disorder; PTSD = posttraumatic stress disorder; SoA = social anxiety; SpF = spider fear; SR = stress responsiveness; StA = state anxiety; TA = trait anxiety.

^a Plus (+) indicates evidence in favor of the idea that changes in attentional bias may lead to changes in fear and anxiety, plus/minus (±) indicates mixed findings, and minus (–) indicates null findings or reversed effects.

students to either an avoid threat group or a no-training control group. The training manipulation extended over eight sessions spread over 3 weeks. Trait anxiety was assessed before and after the training. Although the authors reported that participants in the avoid threat group showed a significant decrease in trait anxiety following the training, the description of the methods, analyses, and results in these experiments is very sparse, which makes it difficult to assess the quality of the experiments and the strength of the data.

More convincing evidence comes from a study by Amir, Beard, Burns, and Bomyea (2009), who assigned participants suffering from clinical GAD to either an avoid threat group or a no-training control group. Following the attentional training manipulation, the avoid threat group showed significant decreases on different self-report measures of anxiety and on a clinician-rated interview, whereas the control group did not show such significant decreases. Furthermore, 50% of the participants in the avoid threat group no longer met the *DSM-IV-TR* (American Psychiatric Association, 2000) criteria for GAD after the training, relative to 13% in the control group.

Similar findings have been reported by Hazen, Vasey, and Schmidt (2009) in a sample of severe worriers. Attentional avoidance training resulted in reduced worrying and reduced feelings of anxiety and depression relative to a no-training control group. However, these findings were only partially replicated by Hayes, Hirsch, and Mathews (2010). These authors also assigned high worriers to either an avoid threat group or a no-training control group and assessed the effect of this training manipulation on mood state and negative thought intrusions during a worry test. After training, participants in the avoid threat group had less negative thought intrusions. However, the groups did not differ with regard to the levels of anxiety or happiness/sadness that were provoked by the worry task. It is possible that the training manipulation was too brief to have a sufficiently strong effect on attentional bias to obtain similar results as Hazen et al. In line with this explanation, Hayes et al. did not find a significant modification of attentional bias as a result of the training, as illustrated by a nonsignificant interaction between attention group (avoid threat vs. control), experiment phase (first part of training vs. second part of training) and congruency (congruent vs. incongruent).

Finally, Krebs, Hirsch, and Mathews (2010) also investigated the effect of attentional training on worrying. They trained moderate worriers either to avoid or to attend to threatening words. In both groups, half the participants were explicitly informed about the training contingencies, whereas the other half received no such information. Results showed that participants in the explicitly instructed avoid threat group had less observer-rated negative thought intrusions than participants in the explicitly instructed attend threat group. However, contrary to the results of Hayes, Hirsch, and Mathews (2010), there were no differences in observer-rated negative intrusions between participants in the attend threat and the avoid threat groups with standard instructions. Furthermore, there were no significant group differences on self-rated negative thought intrusions, nor did the authors report whether the explicitly instructed avoid threat group differed from the attend and avoid threat groups with standard instructions for self-rated or observer-rated negative thought intrusions. Finally, the different training manipulations had no differential effects on either depressive or anxious mood, again illustrating that changes

in attentional bias do not necessarily lead to direct changes in general negative mood.

Several studies have addressed the effects of attentional training on social anxiety. First, Amir, Weber, Beard, Bomyea, and Taylor (2008) showed that socially anxious individuals who were trained to avoid pictures of disgusted faces responded with a smaller increase in state anxiety to a speech task than participants in a no-training control group. The speeches in the avoid threat group were also judged to be of higher quality. Second, S. Li, Tan, Qian, and Liu (2008) found that socially anxious participants who were trained to avoid pictures of threatening faces reported less fear of general social interaction than participants in a no-training control group. However, they did not find significant group differences on two measures of more specific symptoms of social anxiety, such as the fear of being watched or the fear of being evaluated. Third, Amir, Beard, Taylor, et al. (2009) found that participants suffering from clinical social anxiety who were trained to avoid pictures of disgusted faces showed larger decreases in three social anxiety questionnaires than participants in a no-training control group. These gains in decreased social anxiety were maintained at a 4-month follow-up. Of further interest, 50% of the participants in the avoid threat group no longer met the diagnostic criteria of generalized social phobia after the training, compared to only 14% of the participants in the control group. Fourth, N. B. Schmidt, Richey, Buckner, and Timpano (2009) showed that an attentional avoidance training procedure led to reduced levels of clinical social anxiety, as measured by both self-report questionnaires and clinical interviews. Despite these promising early findings, a number of recent studies has failed to replicate these effects (see Table 11).

The experiment of Klumpp and Amir (2010) has yielded a different pattern of results. They assigned moderately socially anxious students to either an attend threat training group, an avoid threat training group, or a no-training control group. As a primary outcome variable, they assessed state anxiety in response to a speech task. In line with Amir et al. (2008), they found that the avoid threat training group responded with a smaller increase in state anxiety than the control group. However, relative to the control group, the attend threat group also showed a smaller increase in state anxiety in response to the speech task, while state anxiety in the attend threat and the avoid threat groups did not differ. Clearly, this finding is unexpected, as it indicates that both attending to threat and attending away from threat may have a beneficial impact on anxiety.

Klumpp and Amir (2010) argued that both the attend and avoid threat training increased participants' levels of attentional control (Derryberry & Reed, 2002), which in turn made them less vulnerable to stress and anxiety. However, this explanation seems premature and requires thorough scientific testing. First, because their experiment did not include a measure of attentional control, it cannot be shown that the training affected the participants' levels of attentional control. Second, their design did not allow for the conventional assessment of changes in attentional bias, making strong claims about these changes impossible. Third, the idea that both attending toward and attending away from threat reduces stress vulnerability is in direct conflict with the original findings of MacLeod et al. (2002), who found a smaller increase in negative affect in an avoid threat group compared to an attend threat group. Fourth, and more fundamentally, questions can be raised about the

attentional control explanation as formulated by Klumpp and Amir. They refer to Derryberry and Reed (2002) when they argue that an increase in attentional control may lead to reduced anxiety vulnerability. However, in the original study of Derryberry and Reed, it was found that anxious individuals with high levels of attentional control had a smaller attentional bias toward threat. Hence, if anything, an increase in attentional control would lead to a decrease in attentional bias. As such, the attentional control hypothesis of Klumpp and Amir still incorporates the hidden assumption that a decrease in attentional bias leads to reduced anxiety vulnerability. It can therefore not explain why an increase in attentional bias would have the same effect.

With regard to spider fear, three studies to date have investigated the effect of attentional bias modification. Harris and Menzies (1998) trained unselected students either to avoid or to attend to spider-related words. They did not find significant differences between the two training groups on a questionnaire of spider fear. This null result is possibly due to the very limited training manipulation. Their training consisted only of 40 trials, which is substantially fewer than that of other training studies (see Table 11). More recently, Reese, McNally, Najmi, and Amir (2010) assigned spider fearful participants to either an attentional avoidance training group or a no-training control group. In line with Harris and Menzies, they also failed to find group differences following training on either self-reported spider fear or behavioral avoidance of a real spider. Van Bockstaele et al. (2011b) trained unselected students either to attend to spiders or to avoid spiders. Although they successfully induced an attentional bias toward spiders in the attend group and attentional avoidance of spiders in the avoid group, the changes in attentional bias were not accompanied by differences between the groups on self-reported spider fear, implicit attitudes toward spiders, physiological fear responses to spiders, or behavioral avoidance of a spider. Hence, within the context of spider fear, three independent studies failed to find effects of attentional manipulations on spider fear.

Najmi and Amir (2010) investigated the effect of attentional bias modification on fear of contamination. They found that participants with elevated levels of OCD who were trained to avoid contamination-related words completed more steps in a behavioral approach task than participants in a no-training control group. However, the groups did not differ with regard to anxiety ratings during the behavioral approach task, nor was there a significant group difference on self-reported symptoms of OCD (N. Amir & S. Najmi, personal communication, March 15, 2011).

In a more theory-driven study, Van Bockstaele, Verschuere, De Houwer, and Crombez (2010) investigated the effect of attentional manipulations on the extinction of a conditioned fear response by training participants to avoid or attend to an acquired signal of threat during an extinction phase. They found that the extinction of the conditioned fear response was faster in the group of participants that was trained to attend to threat compared to both the avoid threat group and a control group in which attention was not trained.

In sum, our overview of attentional bias modification studies suggests that there is relatively coherent evidence for the idea that changes in attentional bias are associated with changes in stress vulnerability (but see Behar et al., 2010), generalized anxiety (but see Krebs et al., 2010), and social anxiety (but see Table 11 for recent null findings). For specific fears, it seems that changes in

attentional bias do not necessarily lead to changes in fear. In some studies, researchers actually tested whether these changes in anxiety were due to changes in attentional bias rather than to changes in some unknown third confounding variable. For example, See et al. (2009) and Amir et al. (2008) showed that the changes in anxiety in their sample were mediated by the induced changes in attentional bias. More modest or partial mediation effects are also reported in the studies of Amir, Beard, Burns, and Bomyea (2009); Amir, Beard, Taylor, et al. (2009); and Najmi and Amir (2010).

In a meta-analysis of 12 studies on attentional training within the context of social anxiety and generalized anxiety, Hakamata et al. (2010, 2012) showed that the modification of attentional bias produced moderately sized changes in anxiety ($d = 0.51$). The fail-safe n in this analysis was 44, indicating that 44 studies reporting null results would be needed to render the effect non-significant. However, after the removal of three possible outlying experiments with very large effects (two experiments by Mathews & MacLeod, 2002, and one experiment by N. B. Schmidt et al., 2009), the size of the effect dropped to 0.36, and the fail-safe n dropped to 7 (Hakamata et al., 2010).

The estimated effect size of cognitive bias modifications on anxiety in the meta-analysis of Hallion and Ruscio (2011) was even more modest, with a Hedges' g (Hedges, 1981) of 0.13 (interpretation of magnitude of Hedges' g is similar to Cohen's d ; see Hallion & Ruscio, 2011). Furthermore, Hallion and Ruscio included the effects of both attentional bias modification procedures and interpretation bias modification procedures (see below), and as such, their estimate does not represent a pure estimate of the effect that attentional training has on fear and anxiety. The pure effect of attention training on measures of anxiety was small and nonsignificant ($g = 0.10$, $p = .13$, 95% CI $[-0.03, 0.22]$; excluding the outlying results of N. B. Schmidt et al., 2009; L. Hallion, personal communication, December 7, 2012). Nevertheless, attentional bias modification did have a statistically reliable effect on emotional responsiveness to stress ($g = 0.31$, $p < .005$, 95% CI $[0.10, 0.51]$). Thus, although several studies have illustrated that the manipulation of attentional bias can have an effect on (mostly self-report) measures of anxiety and responsiveness to stress, the reliability and possible boundary conditions of this effect are not beyond contention. This idea is further supported by a growing number of failures to replicate the finding that attentional bias modification results in changes in anxiety (see Table 11).

A different line of relevant research concerns experiments in which attentional bias is measured before and after participants are randomly assigned to either a control group or a treatment group in which the direct reduction of attentional bias is not specifically targeted (see Table 12). If one can show that therapy reduces anxiety and that this decrease in anxiety is in turn followed by a decrease in attentional bias, this would provide support for the idea that changes in fear or anxiety lead to changes in attentional bias and thus that fear and anxiety are causes of attentional bias rather than the other way around.

Foa and McNally (1986) were among the first to address this issue. In a small sample of OCD patients, they found that attentional bias toward threat-related words was reduced after exposure treatment. However, because they did not include a control group or a measure of effectiveness of the therapeutic intervention, one cannot be sure that the reduction in attentional bias was due to the reduction in fear. More convincing evidence is provided by Watts,

Table 12

Summary of Experimental Evidence: Effects of Therapy on Anxiety and Attentional Bias

Anxiety problem	Study	N	AB measure	Evidence ^a	Outcome
Clinical GAD	Mathews et al. (1995)	18/20	Stroop and visual search	±	Less anxiety and less interference after treatment in both Stroop ($N = 18$) and visual search ($N = 20$), but no no-treatment control group
Clinical GAD	Mogg et al. (1995)	11	Stroop	±	Less symptoms of GAD and less SI after treatment, but no no-treatment control group
Clinical SoA	Pishyar et al. (2008)	32	Dot probe	+	Decrease in SoA and larger decrease in AB after treatment for social threat in treatment group compared to no-treatment control group
Clinical SoA	Baños et al. (2008)	30	Visual search	±	Less SoA and less interference by threatening distractors after treatment, but no no-treatment control group
Clinical SoA	Lundh & Öst (2001)	24	Stroop	±	Less SoA and less SI after treatment in treatment responders, but no no-treatment control group
Clinical SoA	Mattia et al. (1993)	29	Stroop	±	Less SoA and less SI after treatment in treatment responders, but no no-treatment control group
SpF	Watts et al. (1986)	28	Stroop	+	Decrease in SpF and larger decrease in SI in treatment group compared to no-treatment control group
Clinical SpF	Côté & Bouchard (2005)	28	Stroop	±	Less SpF and less SI after treatment, but no no-treatment control group
Clinical SpF	Lavy et al. (1993)	36	Stroop	±	Less SpF and less SI after treatment, but no no-treatment control group
Clinical SpF	van den Hout et al. (1997)	37	Stroop	±	Less SpF and less SI after treatment, but no no-treatment control group
Clinical SpF	Thorpe & Salkovskis (1997)	23	Stroop	–	Decrease in SpF in treatment group, but similar decrease in SI in treatment group and no-treatment control group
Conditioned fear	Koster, Crombez, Van Damme, et al. (2005)	66	Exogenous cueing	±	Decreased US expectancy and decreased AB for conditioned signal of threat following extinction procedure, but no group without extinction procedure
Conditioned fear	Van Damme et al. (2006)	72	Exogenous cueing	±	Decreased AB for conditioned signal of threat following extinction procedure, but no group without extinction procedure and no data on the extinction of fear
PTSD	D'Andrea & Pole (2012)	24	Stroop	±	Marginal decrease in PTSD symptoms and less SI after treatment, but no no-treatment control group
Clinical OCD	Foa & McNally (1986)	9	Dichotic listening task	±	Decreased AB following treatment, but no no-treatment control group and no report on the effectiveness of the treatment

Note. Where applicable, participants in the studies in this table were randomly assigned to different treatment groups. For these studies, we used the following search command on Web of Science: Topic = (attention* bias* AND (fear OR anxiety OR distress OR PTSD OR OCD OR stress OR emotion* OR phobi*) AND ((therap*) OR (treatment) OR (exposure))). The results of this search command were further refined for language (English) and publication year (before December 2012), and we only selected original research articles. Next, we inspected titles and abstracts and removed articles that were not related to fear or anxiety. Finally, we only selected studies in which attentional bias was measured before and after a therapeutic intervention. AB = attentional bias; GAD = generalized anxiety disorder; OCD = obsessive-compulsive disorder; PTSD = posttraumatic stress disorder; SoA = social anxiety; SpF = spider fear; SI = Stroop interference; US = unconditioned stimulus.

^a Plus (+) indicates evidence in favor of the idea that changes in fear and anxiety may lead to changes in AB, plus/minus (±) indicates mixed findings, and minus (–) indicates null findings or reversed effects.

McKenna, Scharrock, and Trezise (1986). They found that the attentional bias of individuals with spider phobia was reduced following therapy compared to individuals with spider phobia in a waiting list control group (but see Thorpe & Salkovskis, 1997). Likewise, Lavy, van den Hout, and Arntz (1993; see also Côté & Bouchard, 2005; van den Hout, Tenney, Huygens, & de Jong, 1997) found that the attentional bias of participants suffering from spider phobia was significantly reduced after successful exposure treatment.

Similar results have also been reported in the context of social anxiety. For instance, Pishyar, Harris, and Menzies (2008) as-

signed socially anxious individuals to either a cognitive behavior treatment group or a waiting list control group. Using the dot probe task, they found that both groups showed an attentional bias toward socially threatening stimuli and attentional avoidance of positive stimuli before treatment/waiting list. Following treatment/waiting list, the treatment group had an attentional bias toward positive stimuli and avoidance of threatening stimuli, whereas the waiting list control group showed no such change (see also Baños et al., 2008; Lundh & Öst, 2001; Mattia, Heimberg, & Hope, 1993). Mathews et al. (1995) showed that attentional bias toward threatening stimuli in GAD patients was reduced following treat-

ment. Furthermore, this reduction in attentional bias was maintained at a 3-month follow-up (see also Mogg, Bradley, Millar, & White, 1995). D'Andrea and Pole (2012) found that participants suffering from PTSD showed reduced Stroop interference by trauma-related words following treatment. Finally, fear conditioning studies have shown that the extinction of a conditioned fear response is associated with the extinction of an attentional bias toward the CS+ (Koster, Crombez, Van Damme, et al., 2004, 2005).

These studies relatively consistently show that attentional bias is reduced after therapy. However, there are a number of notable limitations to this line of research. First, only three of the studies mentioned above included a no-treatment control group (Pishyar et al., 2008; Thorpe & Salkovskis, 1997; Watts et al., 1986). The changes in attentional bias in the majority of the studies that we discussed may therefore reflect a mere effect of practice with the attentional bias measure. Second, none of the studies included the testing of a mediation model in which the change in fear or anxiety mediated the change in attentional bias. One can therefore not exclude the possibility that the therapeutic interventions changed attentional bias and that this change in attentional bias resulted in reduced fear or anxiety (Clark & Beck, 2010).

Despite these limitations, this line of research suggests that attentional bias may be a consequence of fear and anxiety. Nevertheless, it should be noted that this conclusion (i.e., a reduction of fear or anxiety reduces attentional bias; $-B \rightarrow -A$) does not negate the possibility that attentional bias causally influences fear or anxiety (i.e., $A \rightarrow B$). The relevant evidence does render it unlikely that there is a strict unidirectional causal pathway from attentional bias to fear and anxiety. However, the evidence does not preclude the possibility that attentional bias is a cognitive vulnerability factor, or a maintaining or exacerbating factor of fear and anxiety.

Analogy

According to Hill (1965), it is more reasonable to assume that a relation is causal if there is evidence for a causal relation between similar constructs. For instance, if there is evidence for the fact that smoking is causally related to lung cancer, it is also more likely that smoking can cause laryngeal cancer. Of course, finding analogies does not prove causality; it provides only an additional indication that a relation could be causal. Although the analogy criterion provides a less convincing indication for causality than criteria such as temporality and experimental evidence, it remains indicative. Moreover, it allows us to link the literature on the role of attentional bias in fear and anxiety to the literature on other cognitive biases. For instance, there is evidence that other cognitive biases, such as interpretation bias, are causally related to fear and anxiety.

As mentioned before, interpretation bias denotes a tendency to interpret ambiguous stimuli or situations in a threatening way. According to several cognitive models of anxiety, such an interpretation bias is causally related to fear and anxiety (e.g., Beck & Clark, 1997; Mathews & Mackintosh, 1998; Williams et al., 1997). There is indeed ample evidence for the existence of an interpretation bias in different anxiety disorders. For instance, Eysenck, Mogg, May, Richards, and Mathews (1991) found that currently anxious participants were more likely to interpret ambiguous sen-

tences in a threatening fashion, relative to both recovered anxious participants and normal controls (e.g., see also Amir, Beard, & Bower, 2005; Hirsch & Mathews, 1997; MacLeod & Cohen, 1993).

More recently, researchers have begun to investigate the causality of the relation between interpretive bias and fear and anxiety with more scrutiny (see Table 13). Analogous to findings of attention training studies, it has been shown that individuals can be trained to make threatening or benign interpretations of ambiguous stimuli or situations (e.g., Grey & Mathews, 2000). Such benign interpretation training reduces emotional vulnerability to stress (B. Mackintosh, Mathews, Yiend, Ridgeway, & Cook, 2006; Wilson, MacLeod, Mathews, & Rutherford, 2006; but see Saleminck, van den Hout, & Kindt, 2007b), state anxiety (Mathews & Mackintosh, 2000; Yiend, Mackintosh, & Mathews, 2005), trait anxiety (Mathews, Ridgeway, Cook, & Yiend, 2007; Saleminck, van den Hout, & Kindt, 2007c), symptoms of GAD (Hayes, Hirsch, Krebs, & Mathews, 2010; Hirsch, Hayes, & Mathews, 2009), and symptoms of social anxiety (e.g., Beard & Amir, 2008; R. Murphy, Hirsch, Mathews, Smith, & Clark, 2007). Interestingly, consistent with the attention retraining studies, benign interpretation training has no effect on spider fear (Teachman & Addison, 2008). Taken together, these studies suggest that there is a causal relation between interpretive biases and anxiety, which makes a similar causal relation between attentional bias and anxiety conceivable.

Discussion and Conclusions

Many cognitive theories postulate that attentional bias toward threat is causally involved in the etiology, maintenance, or exacerbation of fear and anxiety. In this review, we investigated the empirical support for these claims. More specifically, we examined whether the available evidence about the relation between attentional bias and fear and anxiety meets Hill's (1965) criteria of causation. Although an evaluation of Hill's criteria does not allow for definite conclusions about causality, they do provide a useful guide for selecting and organizing a wide range of studies that are relevant to answer the causality question.

Our review shows that there is convincing evidence for a small to moderately sized relation between attentional bias toward threat and fear and anxiety (strength of the association). However, there are important boundary conditions to this relation, as illustrated by a considerable degree of inconsistency among different studies that used different methodologies and populations. There is only limited evidence for the existence of a dose-response curve, as evidenced by the relatively small and the inconsistent pattern of correlations between attentional bias and measures of fear and anxiety. Although a plausible mechanism has been described through which attentional bias may lead to increased fear and anxiety, other theories assume that attention for threat should reduce fear and anxiety rather than promote it.

Prospective studies have shown that attentional bias is predictive of later stress vulnerability and fear responses, and thus that attentional bias can precede the onset of fear and anxiety in time (temporality). However, the absence of large-scale longitudinal studies, in which attentional bias at Time 1 is used to predict fear or anxiety disorders at Time 2, constitutes a major gap in the current state of the literature. Also, the finding that fear or anxiety may precede the onset of attentional bias toward threat is not

Table 13

Summary of Evidence for a Causal Relation Between Interpretation Bias and Fear and Anxiety

Anxiety problem	Study	N	Groups	Evidence ^a	Outcome
SR	Wilson et al. (2006)	44	Benign, negative	+	Increased negative mood in negative group following stressor; no changes in benign training group
SR, StA	B. Mackintosh et al. (2006), Experiment 2	40	Benign, negative	+	Increased StA in negative group following stressor; no change in StA in benign training group
SR, StA, TA	Salemink et al. (2007b)	118	Benign, negative	–	No effects of interpretation training on StA or negative mood in response to stressor; short-lived increase in TA in negative group
StA	Mathews & Mackintosh (2000), Experiment 1	20	Benign, negative	+	Increased StA in negative group
StA	Mathews & Mackintosh (2000), Experiment 4	54	Benign, negative	±	Increased StA in negative group and decreased StA in benign training group, but only when interval between training and anxiety measurement was short
StA	Mathews & Mackintosh (2000), Experiment 5	56	Benign, negative	±	Increased StA in negative group and decreased StA in benign training group, but only when participants generated the interpretations themselves
StA	B. Mackintosh et al. (2006), Experiment 1	48	Benign, negative	–	No group differences in StA
StA	Mathews & Mackintosh (2000), Experiment 2	20	Benign, negative	–	No group differences in StA
StA, TA	Salemink et al. (2007c)	81	Benign, negative	+	Reduced StA and larger decrease in TA in benign training group
StA, TA	Yiend et al. (2005), Experiment 1	20	Benign, negative	±	Decreased StA in benign training group, but no effect on TA
StA, TA	Yiend et al. (2005), Experiment 2	24	Benign, negative	–	No group differences in either TA or StA
StA, TA	Yiend et al. (2005), Experiment 3	19	Benign, negative	–	No group differences in either TA or StA
TA	Mathews et al. (2007)	39	Benign, control	+	Stronger decrease in TA in benign training group
GAD/worry	Hirsch et al. (2009)	40	Benign, control	+	Less negative thought intrusions and less anxiety in benign training group
Clinical GAD	Hayes, Hirsch, Krebs, and Mathews (2010)	40	Benign, control	±	Less negative thought intrusions in benign training group, but no effect of training on mood
SoA	Beard & Amir (2008)	27	Benign, control	+	Larger decrease in SoA in benign training group
SoA	R. Murphy et al. (2007)	66	Benign, control	±	Less anticipated anxiety in response to upcoming social situation in benign group, but no group differences in StA
SpF	Teachman & Addison (2008)	61	Benign, control, no training	–	No group differences in avoidance of a real spider and anxiety ratings

Note. Participants in the studies in this table were randomly assigned to different training groups. For these studies, we used the following search command on Web of Science: Topic = (interpret* bias* AND (fear OR anxiety OR distress OR PTSD OR OCD OR stress OR emotion* OR phobi*) AND ((train*) OR (manipulate*) OR (modification) OR (IBM))). The results of this search command were further refined for language (English) and publication year (before May 2011), and we only selected original research articles. Next, we inspected titles and abstracts and removed articles that were not related to fear or anxiety. Finally, we only selected studies in which interpretation bias was manipulated in one group and changes in fear or anxiety were compared with a control group. GAD = generalized anxiety disorder; SoA = social anxiety; SpF = spider fear; SR = stress responsiveness; StA = state anxiety; TA = trait anxiety.

^a Plus (+) indicates evidence in favor of the idea that changes in interpretation bias may lead to changes in fear and anxiety, plus/minus (±) indicates mixed findings, and minus (–) indicates null findings or reversed effects.

compatible with the idea that attentional bias always precedes fear or anxiety in time.

The strongest evidence for causality comes from a recent line of experimental studies. These studies have shown that experimentally induced changes in attentional bias can causally influence stress reactivity, symptoms of GAD, and symptoms of social anxiety (but see Hallion & Ruscio, 2011). However, research on the effectiveness of such attentional bias modification procedures for the reduction of specific fears (e.g., spider fear) has not yielded any significant results, suggesting that the distinction between fear and anxiety may be an important boundary condition. Also, the finding that reductions in fear and anxiety diminish attentional bias can be regarded as an indication that attentional bias at least sometimes is a consequence rather than a cause of anxiety (see Mogg & Bradley, 1998). Nevertheless, we have argued that this evidence does not preclude the possibility that attentional bias toward threat is a cognitive vulnerability factor or maintaining

factor of fear and anxiety. Finally, there is evidence for an analogously causal relation between interpretation bias and anxiety.

Our review of the literature suggests that attentional bias toward threat is probably more than an epiphenomenon of fear and anxiety. A crucial issue concerns the direction of causality. If attentional bias is a mere epiphenomenon of fear and anxiety, one should be able to find evidence for the impact of fear and anxiety on attentional bias but not the other way around. Several criteria (e.g., strength of the relation, consistency, dose-response curve) cannot be used to differentiate between the possibility that attentional bias causes fear or anxiety and the possibility that fear and anxiety cause attentional bias. Several prospective studies provide evidence for the idea that attentional bias precedes fear and anxiety in time (see Temporality). Crucially, experimental manipulations of attentional bias do—at least under certain conditions—lead to changes in fear and anxiety (see Experiments). These results indicate that attentional bias is not merely an epiphenomenon of

fear and anxiety and that attentional bias causally contributes to fear and anxiety. However, the finding that reductions in fear or anxiety may lead to reduced attentional bias does suggest that fear and anxiety may be a cause of attentional bias. Similarly, the conditioning results that we have discussed in the section on temporality offer some evidence for the idea that fear and anxiety may precede the development of attentional bias in time, again indicating that fear and anxiety may be a cause of attentional bias.

The evidence suggests that attentional bias may be a cognitive vulnerability factor for the development of fear and anxiety (Pérez-Edgar et al., 2010, 2011). However, given the evidence suggesting that fear and anxiety are causes of attentional bias and the evidence for the idea that attentional bias is a cause of fear and anxiety (see Temporality and Experiments), the relation between attentional bias and fear and anxiety is probably best defined as a bidirectional relation, with attentional bias maintaining or exacerbating fear and anxiety, but also with fear and anxiety maintaining or exacerbating attentional bias.

These considerations have important implications for theories of fear and anxiety that we discussed at the start of our article. First, the model of Bar-Haim et al. (2007) postulates that attentional bias is unidirectionally involved in the etiology of fear and anxiety, or that attentional bias is at least a cognitive vulnerability factor for fear or anxiety. As discussed above, our review provides only limited support for this proposal. Second, we did not find strong support for the causal claims in the cognitive-motivational view of Mogg and Bradley (1998). These authors argued that the pattern of initial vigilance and subsequent avoidance is at best involved only in the maintenance of anxiety. However, several attentional retraining studies (see Experiments) illustrate that the increase of attentional avoidance of threat leads to lower levels of anxiety rather than maintaining anxiety. Third, the model of Beck and Clark (1997) is difficult to evaluate in the light of the evidence that we reviewed because the model is not very specific. On the basis of their theoretical work, one could argue that (a) attentional bias and anxiety are causally unrelated correlates or (b) attentional bias is a link in a causal chain leading to anxiety. Whichever interpretation one adheres to, neither seems to be supported by the evidence. On the one hand, evidence in support of a causal effect of attentional bias on fear and anxiety contradicts the idea of a spurious correlation. On the other hand, the chain idea implies a unidirectional cause–effect model that is not compatible with the observed effects of fear and anxiety on attentional bias. The evidence that we reviewed in the present article fits best with the theorizing of Williams et al. (1997) and Eysenck (1997), who assumed a mutually reinforcing relation between attentional bias toward threat and fear or anxiety. Furthermore, attentional bias is in both these models also considered a cognitive vulnerability factor for the development of fear or anxiety (Pérez-Edgar et al., 2010, 2011).

The circularly causal relation between attentional bias and fear and anxiety and especially the malleability of attentional bias may have clinical implications. The recent line of research on attentional bias modification suggests that training anxious individuals to attend away from threat can lead to reductions in anxiety (Hakamata et al., 2010, but see also Hallion & Ruscio, 2011). An advantage of these attentional training programs is their ease of administration and hence their low costs for the individual, in both time and money. Such new approaches to clinical treatment may prove important to lighten the burden of mental health problems (Kazdin & Blase, 2011).

However, at least three issues should be considered before such attentional training procedures are disseminated and implemented in clinical practice. First, it is unclear whether attentional bias modifications have a beneficial impact on all types of anxiety disorders. For instance, there is no convincing evidence that changes in attentional bias have an impact on specific fears (Harris & Menzies, 1998; Reese et al., 2010; Van Bockstaele et al., 2011b). It is possible that attentional training may help to reduce the more general and enduring feelings of unrest and uncertainty that mark trait anxiety, general anxiety, and social anxiety, but less so to reduce the intense and well-circumscribed feelings of immediate fear (Barlow, 1988; Craske & Waters, 2005; Rachman, 1998).

Second, there is little research on the potentially harmful effects of attentional avoidance training. Training individuals to attend away from threat may hamper habituation during exposure (Foa & Kozak, 1986; see also Van Bockstaele et al., 2010) and can be considered cognitive avoidance, which has been argued to maintain anxiety (e.g., Dugas, Gagnon, Ladouceur, & Freeston, 1998; Steil & Ehlers, 2000). Also, more research is needed to fully understand which individuals may benefit from such attentional training programs. For instance, it is unclear whether attention training is beneficial only for anxious individuals with an attentional bias toward threat or also for anxious individuals without attentional bias. It is also not yet clear whether attention training is effective for the prevention of fear and anxiety in currently nonanxious individuals who have an attentional bias toward threat.

A third issue concerns the size of the therapeutic effects of attentional training. It remains to be investigated whether the therapeutic effects of attentional training are large enough to justify the use of attentional bias modification as a stand-alone treatment, or whether it should be incorporated in or combined with traditional therapies for fear and anxiety. These three issues as well as a recent number of null findings in the field of attention training (e.g., Boettcher et al., 2012; Bunnell, Beidel, & Mesa, 2013; Carlbring et al., 2012; Julian, Beard, Schmidt, Powers, & Smits, 2012; Neubauer et al., 2013; Schoorl, Putman, & Van Der Does, 2013; see Table 11) should be taken into consideration when implementing attentional training in clinical practice. The commercialization of the attentional training procedure, which is advertised as a successful stand-alone means to reduce anxiety (see <http://www.managingyouranxiety.com>), seems at least premature (see also Emmelkamp, 2012).

It is important to consider also the limitations of our analysis. Our review of the literature was organized in terms of a subset of Hill's (1965) criteria for causation. This allowed us to present a broad and structured overview of all the available evidence that is relevant for assessing the causal nature of the relation between attentional bias and fear and anxiety. We wish to repeat, however, that evaluating Hill's criteria (or any other criteria for causality, for that matter) does not allow one to draw definite conclusions regarding issues of causality. There is a long-standing debate among philosophers of science and epidemiologists on whether and how Hill's criteria should be used and what their value is for the study of causality (Rothman & Greenland, 2005; Susser, 1991; Ward, 2009a, 2009b). This debate led to several conclusions.

First, it is clear that any conclusion based on the criteria contains a level of uncertainty and subjectivity. There is no objective manner to determine whether evidence in favor of a certain criterion is sufficient, or whether certain evidence outweighs possible conflicting evidence. There also is no objective standard by which one can decide whether

convincing evidence for certain criteria allows one to infer causation, whether such evidence is needed in favor of every single criterion, or whether the falsification of a single criterion suffices to reject all causal claims (Ward, 2009b). Second, Hill's (1965) causal criteria are not fully compatible with more complex forms of causation (for detailed discussions, see Höfler, 2005; Ward, 2009a). Because we organized our review on the basis of Hill's criteria, we did not take into account interactions between different possible causes of fear and anxiety or complex causal pathways in which attentional bias toward threatening stimuli mediates or moderates the impact of other causes.

Another limitation of our analysis concerns the fact that some of Hill's (1965) criteria assume a *monotonic linear relation* between cause and effect. For instance, this is the case for the dose-response criterion. This linear approach is in line with some models of attentional bias in fear and anxiety (e.g., Williams et al., 1997), but there is at present no consensus as to whether the relation between attentional bias and fear and anxiety is indeed linear (e.g., Mogg & Bradley, 1998).

Finally, it is important to note that our study did not apply strategies of systematic reviews or meta-analysis. Instead, for seven of Hill's (1965) criteria, we selected and described the literature that seemed most relevant for a particular criterion. Although we tried to be comprehensive, it was sometimes not possible to formulate stringent search terms for each criterion. It is therefore possible that there are some blind spots in our selection of literature. Nevertheless, Hill's criteria provided us with useful guidelines for the selecting and organizing the available evidence, almost as a "road map through complicated territory" (Rothman & Greenland, 2005, p. S148). It is exactly this heuristic value of Hill's criteria that we exploited in the present article.

Despite these limitations, our review of the available literature does support the conclusion that attentional bias is causally related to fear and anxiety. Although it is unlikely that attentional bias leads to fear and anxiety in a strict unidirectional manner, we consider the relation between attentional bias and fear and anxiety to be bidirectional, maintaining, or exacerbating. Given this causal influence, a fine-grained understanding of the effects of attentional factors seems fruitful in both understanding and modifying anxiety.

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(Appendices follow)

Appendix A

List of Studies With the Emotional Stroop Paradigm, Not Included in Table 3

Reference	Reason for not including the article in Table 3
Benoit et al. (2007)	Sample consists of children and adolescents
Charash & McKay (2009)	Deals with disgust rather than fear/anxiety
Dresler et al. (2009)	No anxious versus nonanxious groups
Frühholz et al. (2011)	No anxious versus nonanxious groups
Hadwin et al. (2009)	Sample consists of children; no straightforward between-group analyses
Jansson & Lundh (2006)	No straightforward between-group comparisons
McKay (2005)	No straightforward between-group comparisons; necessary follow-up analyses not reported
Mueller-Pfeiffer et al. (2010)	Paradigm deviates from our description
Oathes et al. (2011)	Paradigm deviates from our description
Richards et al. (2007)	Sample consists of children
Sveen et al. (2009)	No anxious versus nonanxious groups
Taake et al. (2009)	No between-group comparisons
Thomas et al. (2007)	No anxious versus nonanxious groups
van Peer et al. (2010)	No anxious versus nonanxious groups
Vendemia & Rodriguez (2010)	Paradigm deviates from our description
Vythilingam et al. (2007)	Paradigm deviates from our description

Appendix B

List of Studies With the Dot Probe Task, Not Included in Table 4

Reference	Reason for not including the article in Table 4
Bannerman et al. (2009)	No anxious versus nonanxious groups
Buodo et al. (2010)	Paradigm deviates from our description
Carlson & Reinke (2008)	No anxious versus nonanxious groups
Cooper & Langton (2006)	No anxious versus nonanxious groups
Dodd & Porter (2011)	Sample consists of patients with Williams syndrome, not a genuine anxiety disorder
Fani et al. (2011)	No anxious versus nonanxious groups
Fawzy et al. (2006)	No analyses of attentional bias scores
Gamble & Rapee (2010)	No reaction time data
Heim-Dreger et al. (2006)	No between-group comparisons; sample consists of children
Helzer et al. (2009)	Sample consists of children
Holmes et al. (2009)	No anxious versus nonanxious groups
Hunt et al. (2006)	Sample consists of children
Kallen et al. (2007)	Sample consists of children
Klumpp & Amir (2009)	No analyses of attentional bias scores
Lee & Knight (2009)	Analyses are too limited
Lonigan & Vasey (2009)	Sample consists of children
MacNamara & Hajcak (2010)	Paradigm deviates from our description
McKay (2005)	Paradigm deviates from our description
Miyazawa & Iwasaki (2009)	No anxious versus nonanxious groups
Mor & Inbar (2009)	No between-group comparisons
S. E. Murphy et al. (2008)	No anxious versus nonanxious groups
Roy et al. (2008)	Sample consists of children
Sarlo & Munafò (2010)	Paradigm deviates from our description
Schrooten & Smulders (2010)	Paradigm deviates from our description
Sutton & Altarriba (2011)	No anxious versus nonanxious groups
Tan et al. (2011)	Paradigm deviates from our description
Telzer et al. (2008)	Sample consists of children, no anxious versus nonanxious groups
Vrijssen et al. (2009)	Paradigm deviates from our description
Waters, Henry, et al. (2010)	Sample consists of children
Waters, Kokkoris, et al. (2010)	Sample consists of children
Waters, Mogg, et al. (2008)	Sample consists of children
Waters, Wharton, et al. (2008)	Sample consists of children

(Appendices continue)

Appendix C

List of Studies With a Cueing Paradigm, Not Included in Table 5

Reference	Reason for not including the article in Table 5
Bannerman et al. (2010a)	No between-group comparisons
Bannerman et al. (2010b)	No between-group comparisons
Bar-Haim et al. (2005)	Paradigm deviates from our description
Brühl et al. (2011)	Paradigm deviates from our description
Cisler et al. (2011)	No between-group comparisons
Cisler & Olatunji (2010)	No analyses of attentional bias scores
Cooper et al. (2009)	No straightforward between-group comparisons
Ellenbogen & Schwartzman (2009)	No analyses of attentional bias scores
Fox et al. (2008)	No analyses of behavioral data
X. Li et al. (2007)	No analyses of attentional bias scores
Maril et al. (2007)	No measurement of attentional bias toward threat (no threatening stimuli)
McTeague et al. (2011)	Paradigm deviates from our description
Mercado et al. (2009)	Paradigm deviates from our description
Moritz et al. (2009)	Paradigm deviates from our description
Moriya & Tanno (2009)	No measurement of attentional bias toward threat (no threatening stimuli)
Okon-Singer et al. (2011)	Paradigm deviates from our description
Pérez-Dueñas et al. (2009)	Paradigm deviates from our description
Van Damme et al. (2008)	No between-group comparisons
Van Damme & Crombez (2009)	No between-group comparisons
Verkuil et al. (2009)	Paradigm deviates from our description
Waters et al. (2007)	No analyses of attentional bias scores
Webb et al. (2010)	Paradigm deviates from our description

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